

Limits to the learnability of abstract underlying representations in Maga Rukai

Abstract. Standard morphophonological analysis allows for **composite** Underlying Representations (URs), which combine information from multiple allomorphs and are abstract in that they do not correspond to any single allomorph. This paper probes at the learnability of composite URs by looking at diachronic restructuring in the Maga dialect of Rukai. In Maga, rhythmic vowel deletion makes it so that no single allomorph has all the underlying vowels, and URs are composite (e.g. [t_{mu}₂su]~[ik-ti₁msu:] → /ti₁mu₂su/). Drawing on corpus and comparative data, I find that paradigms have i) undergone substantial leveling, and ii) extended a vowel-matching alternation. Both changes make surface forms predictable from each other, removing the need for abstraction. To substantiate these results, I implement a maximum entropy model of Maga vowel alternations, and show that a model with concrete URs (that correspond to surface allomorphs) outperforms an abstract-UR model at predicting the patterns of restructuring found in Maga.

1 Introduction

In morphophonemic analyses, alternations are explained by positing abstract underlying representations (URs), with rules or constraints that derive the surface forms. Once we allow URs to deviate from surface forms, we must also ask how abstract URs can be. In other words, how much can URs diverge from their surface representations? This paper argues that abstract URs are dispreferred by learners, drawing on empirical evidence from Maga Rukai, which has undergone restructuring in a way that turned an abstract-amenable analysis into a more concrete one.

Early work in generative phonology, centered around SPE (Chomsky & Halle 1968), allowed for high degrees of abstractness in underlying forms. Since then, various scholars have questioned how abstract URs can be (e.g. Kiparsky 1968, Derwing 1974, Tranel 1981); this line of work argues that speakers cannot learn overly abstract URs, and morphophonological patterns which require positing highly abstract URs will either be unproductive or relearned as more concrete patterns. A more moderate view, along the same lines, is that learners prefer concrete patterns, and only learn abstract representations given enough evidence (e.g. Belth 2023); alternatively, learners may permit highly abstract URs, but doing so increases learning difficulty, and consequently increases the likelihood of acquisition errors (Wang & Hayes 2025).

Much of the empirical evidence against abstract URs comes from language change. Kiparsky's (1968, 1973, 1982) work was for utilizing data from language change to highlight possible limits on abstractness. For example, Kiparsky 1973: p. 27-28 finds that in Mongolian, the sound change of *u>i resulted in stems with surface [i] that take back harmony. However, this pattern subsequently evolved so that surface [i]'s uniformly take front harmony, removing the need to posit abstract underlying /u/. Similar types of "concretization" have been found for Yowlumne (Yawelmani) (Hansson & Sprouse 1999) and Polish jer alternations (Jarosz 2008, Rysling 2016),

among other languages. Notably, most of this body of research has focused on the learnability of abstract phonemes (which never surface). Other more moderate levels of abstraction are not as well-understood, though some recent work suggests that even these patterns can be restructured in a more concrete direction (e.g. Seediq; Kuo 2023a).

Maga Rukai (henceforth Maga) is an Austronesian language spoken in Southern Taiwan (Kaohsiung City) (Li 1977b). Maga is particularly suitable for addressing issues around representational abstractness, because it exhibits morphophonemic alternations that do not require abstract phonemes, but still do require positing relatively abstract URs. The pattern of interest is rhythmic syncope (vowel deletion); the actual environment for syncope is slightly more complicated, but for now it is adequate to say that every odd medial vowel counting from the left is deleted (Chen 2008).¹ Representative examples are given in (1), using the unaffixed stem and negative forms. In these examples, the deleted vowel is indicated with an underscore. The negative prefix /i-/ shifts which vowels get deleted, resulting in $\emptyset \sim V$ alternations between the stem and negative allomorphs.

(1) *Examples of $\emptyset \sim V$ alternations in Maga*

	UR	STEM	NEGATIVE	GLOSS
a.	/kupici/	k_píci	i-kup_cí:	‘sting (NEG)’
b.	/kulubuju/	k_lub_nu	i-kul_buɲú:	‘tortoise (NEG)’

As a consequence of rhythmic syncope, many words require positing **composite URs**, which are abstract in the sense that the UR is not identical to any one surface allomorph, but must instead combine information from multiple allomorphs. For example, ‘sting’ (1a) would typically be analyzed as having the UR /kupici/, which takes its initial vowel from the negative form and its second vowel from the stem form.

The Maga syncope pattern is likely to be unstable and prone to restructuring for reasons discussed in Section 3; in brief, it is opaque and not phonologically optimizing. Observing how restructuring has happened in Maga can therefore be informative for understanding the degree to which learners maintain abstract URs. That is, does restructuring ever happen in a way that preserves the underlying vowel contrasts? Or, does it tend to preserve contrasts present in only one surface allomorph, implying that learners have acquired a non-composite UR? Kiparsky (1973:26-27) notes that in historical change, abstract URs are never “resurrected” as surface pronunciations, and uses this as an argument against abstract representations. By this logic, if Maga speakers mislearn paradigms in such a way that consistently results in non-composite URs, this would suggest that composite URs are dispreferred.

In this paper, I argue that the Maga alternation pattern, while likely to have been phonologically motivated at some point, has been restructured into a more concrete but also morphologically

¹Hsin (2000) characterizes deletion as first building iambic feet from left to right, followed by deletion of the initial vowel of each foot. Li (1977b) characterizes the pattern as deletion of even-numbered vowels counting from the stressed vowel, but as discussed by Hsin, this approach complicates the analysis of stress assignment.

motivated pattern. I track how the Maga syncope pattern has been restructured over time, by comparing Proto-Rukai to modern Maga data from Hsin (2000). To preview the results, I find that a large proportion of ‘composite UR’ paradigms in Maga have been restructured to result in more concrete URs. Two patterns of restructuring are observed: first, some paradigms have undergone so-called **vowel-matching** changes, which maintain the surface vowel alternation pattern but make paradigms less abstract. Other forms have been **leveled**, resulting in a loss of $\emptyset \sim V$ alternations. To explain these changes, I propose that rather than learning a rhythmic syncope pattern, speakers have instead learned a system of morphologically-conditioned vowel alternations.

The rest of this paper is organized as follows: Section 2 describes the Maga rhythmic syncope pattern. Section 3 highlights potential challenges to the learnability of Maga syncope as a phonological pattern. Section 4 formalizes UR abstractness along a hierarchy and places Maga in this hierarchy. In Section 5, I show that two main types of paradigm restructuring have occurred: paradigm leveling (5.1) and vowel matching (Section 5.3-5.5).

In Section 6, I move towards a concrete analysis of Maga vowel alternations. I first show that the pattern appears to be morphologically conditioned, and that even existing metrical analyses of Maga require special reference to morpheme identity. I then implement a concrete analysis where Maga alternations are motivated by an indexed morpheme constraint, and are not phonologically optimizing. This model is shown to be better at predicting the Maga restructuring patterns than an abstract UR model. Finally, Sections 7-8 end with a discussion of the broader implications of these results for our understanding of abstractness.

2 Rhythmic syncope in Maga Rukai

Maga is the dialect of Rukai spoken in Maolin District, primarily in Maolin village within that district (Li 1977a). The population of the village was reported to be 856 in 2024 (Maolin District Office 2025), but the number of fluent speakers is much less than that. Maga is the name of the village used by Chinese and Japanese people during the Japanese occupation of Taiwan (Chen 2008). The local community identify themselves and the dialect by the name *tildika* (Li 1977a), but I use the term Maga to be consistent with prior literature.

Earlier descriptive work on the Maga dialect includes Li (1975), who mainly describes the phonology and morphophonology, as well as Saillard (1995, 1997), who describes the morphology and syntax of Maga. General cross-comparative work on the Rukai dialects includes Tsuchida (1975), who provides wordlists and phonetic descriptions of the Rukai dialects. Additionally, Li (1977b) lays out an internal subgrouping for the Rukai dialects, reconstructs the phonological system of Proto-Rukai, and provides a list of around 700 cognates from dialects of Rukai; around 500 of these cognate sets include Maga. Subsequently, Zeitoun (1995, 2024) provided cross-dialectal overviews of Rukai phonology and morphosyntax.

Hsin (2000, 2003) and Chen (2008) have done the most in-depth work on Maga phonology, providing both documentation and formal analyses of the rhythmic syncope pattern. In partic-

ular, Hsin (2000) collected the data which is the basis of the current paper, and puts forth a well-motivated cyclic phonology analysis of the Maga morphophonemic vowel alternations. Subsequent work by Chen (2008) focuses on analyzing rhythmic syncope in parallel OT; note that Chen focuses on analyzing the pattern in monomorphemes, rather than alternations triggered by affixation. Both Hsin (2000) and Chen (2008) focus on formalizing the rhythmic syncope alternation; they acknowledge exceptions to the syncope pattern, but these are explained via mechanisms such as diacritic marking. The focus of the current paper is to instead explain the large number of exceptions to regular syncope in Maga paradigms, and show that such exceptions are systematic.

In the rest of this paper, examples and descriptive generalizations are taken from Hsin (2000), supplemented with data from Li (1975). The resulting corpus has 1049 stems; of these, a subset of 790 words are given in both their stem and prefixed forms, allowing us to observe the syncope pattern.²

2.1 Basics of Maga phonology and morphology

Maga has seven vowels, /i e u o i ə a/, as well as the consonants in Table 1. The two central vowels /i/ and /ə/ are marginally contrastive; while there are minimal pairs between the two vowels (e.g. [biki] ‘pig’ vs. [bəki] ‘nose running’), [ə] is sometimes an allophone of /i/, and the perceptual distinction between them is unclear, even for native speakers (Hsin 2000: p. 37).

	Labial	Dental	Alveolar	Retroflex	Dorsal
Stop	p b		t d	ɖ	k g
Affricate			c [tʂ]		
Fricative	v	θ (ð)	s (z)		
Nasal	m	n		ɳ	
Lateral			l		
Trill			r~r		

Table 1: Consonant inventory of Maga Rukai

Stress in Maga is argued to be predictable, falling on either the penultimate syllable (e.g. [ábu] ‘ash’) or on final heavy (C)V: syllables (e.g. [adó:] ‘mud’). Maga words never have final consonants, and both initial and medial consonant clusters are restricted to two segments. Hsin (2000) and Chen (2008) both analyze Maga syllables as having a (C)(C)V structure, where no codas are allowed and medial clusters are analyzed as complex onsets (e.g. [blá.vni] ‘silver’).

In terms of the morphology, most verbs are bound roots and obligatorily appear with either the active verb marker /u-/ or the stative verb marker /ma-/ in the citation form (e.g. 2a-b). Nouns can exist as bare stems (e.g. 2c). In this paper, I focus primarily on comparisons between the stem and negative forms, because negation is well-attested and found across all lexical categories. The stem and negative forms also represent the range of possible syncope patterns within

²Note that there are some inconsistencies between Li (1975, 1997) and Hsin (2000) in the transcription of stop voicing. For consistency, I use the transcription provided in Hsin (2000).

a paradigm. Negation is formed using the affix /i- -V/, which has both a prefixal component and final lengthening of the root. Both nouns and verbs can take the negative prefix /i-/, but stative verbs and nouns must take an additional stative prefix /ka-/ [k-]. Note that when the root starts with [k], the stative prefix does not surface, as in (2d). Additionally, while the stative prefix is analyzed as underlyingly /ka-/, it almost always surfaces as [k-] due to vowel syncope.

(2) *Examples of Maga stem-negative paradigms*

	TYPE	STEM	NEGATIVE	GLOSS
a.	Active verb	u-cɲili:	i-ciɲli:	‘see’
b.	Stative verb	ma-rgí:	i-k-ragí:	‘good’
c.	Noun	ɖmári	i-k-ɖamri:	‘moon’
d.	k-initial stative verb/noun	kpáɲi	i-kapɲí:	‘footprint’

Example (3) below shows how the negative prefix can be used in sentential context. Saillard (1995) describes the negative prefix as having narrow scope; sentential negation is indicated using different mechanisms.

(3) *Example of negation in Maga (Saillard 1995: p. 49)*

a.	ma-lcuru	kiki	b.	i-k-lucurú:	kiki
	STAT-FAT	1SG.NOM		NEG-STAT-fat	1SG.NOM
	‘I am fat’			‘I am not fat.’	

2.2 Basic syncope pattern

In Maga isolation stems, odd medial vowels counting from the left are deleted. This type of rhythmic syncope, or the deletion of vowels in metrically weak positions, has been observed in languages like Macushi (Hawkins 1950, Kager 1997), Nishnaabemwin (Valentine 2001, Bowers 2012), and Southern Pomo (Kaplan 2022). The examples in (4), where syllable counts are included for clarity, illustrate the syncope pattern for isolation stems. Note that in stems where syncope results in a monosyllabic surface form, the remaining vowel lengthens (e.g. [rná:], cf. *[rná]); this can be attributed to a typologically well-attested word-minimality restriction, requiring that content words be minimally bimoraic (Hayes 1995).

(4) *Examples of syncope in bare stems*

UR	SR	gloss
/ra ₁ na ₂ /	[r_ná:]	‘creek’
/ɖa ₁ ma ₂ ri ₃ /	[ɖ_mári]	‘moon’
/ri ₁ gi ₂ ri ₃ gi ₄ /	[r_gír_gi]	‘mountain’
/ta ₁ ka ₂ su ₃ lu ₄ ɖu ₅ /	[t_kas_lúɖu]	‘shrimp’

In complex words, however, syncope does not always appear to apply to odd-numbered vowels. Historically and in the synchronic presumed URs, Maga words have no consonant clusters. Consonant clusters on the surface, then, must result from syncope. As shown in (5), this means that each surface consonant cluster, shown in bold, can be used to infer the position of a deleted vowel. By this logic, negative forms of active verbs like (5a) straightforwardly have syncope of odd-numbered vowels. However, in negative forms like (5b), which take the stative prefix, syncope now appears to apply to even-numbered vowels. In other forms such as the imperative, indicated by the suffix */-a:/*, syncope also appears to apply to even-numbered vowels.

(5) *Syncope in complex words*

Type	SR	Position of vowels	gloss
a. /i-/ ‘NEG’ + stem	[i- ciŋli:]	[i ₁ -ci ₂ ŋ ₃ li ₄]	‘not see’
b. /i-/ ‘NEG’ + /k-/ ‘STAT’ + stem	[i-k- ɖamri:]	[i ₁ -k ₂ -ɖa ₃ m ₄ ri ₅]	‘not moon’
c. stem + /-a/ ‘IMP’	[ciŋl-á:]	[ci ₁ -ŋ ₂ l ₃ -á ₃]	‘see! (imp)’

To explain this, Hsin proposes that the domain of syncope is not the entire word, but rather the base that the outermost affix attaches to. Additionally, negative forms of active verbs actually have an input like */i-u-ciŋili-V/*, with the active verb prefix */u-/*. Likewise, imperative forms underlyingly have the active verb prefix, e.g. */u-ciŋili-a/*. The derivations in (6) demonstrate how these assumptions can be used to derive the correct output; parentheses are used to mark the domain of syncope. Given the input */i-ka-ɖamari/*, the domain of syncope is [ka₁ɖa₂ma₃ri₄], and deletion of odd-number vowels results in the correct output [i-k-ɖamri:]. For both negative forms of active verbs and imperatives, the domain of syncope is [u-ciŋili], with the */u-/* prefix. As a result, when syncope applies, */u-/* deletes, and the second vowel of the root deletes, giving us the correct output.

(6) *Domain of syncope in simple vs complex words*

	Nouns and stative verbs		Active verbs		
	a.	b.	c.	d.	e.
gloss	‘moon’	‘neg-stative-moon’	‘act-see’	‘neg-active-see’	‘see-imp’
UR	/ɖamari/	/i-ka-ɖamari/	/u-ciŋili/	/i-u-ciŋili-V/	/u-ciŋili-a:/
syncope	(ɖ ₁ ma ₂ ri ₃)	i-(k ₁ -ɖa ₂ m ₃ ri ₄)-V	u-(c ₁ ŋi ₂ li ₃)	i-(₁ -ci ₂ ŋ ₃ li ₄)-V	(₁ -ci ₂ ŋ ₃ l)-á:
SR	[ɖmari]	[i-k-ɖamri:]	u - [cŋili]	[i-ciŋli:]	[ciŋl-á:]

More examples of alternations between the stem and negative forms are given in (7).

(7) *Examples of rhythmic syncope in Maga*

	UR	STEM	NEGATIVE	GLOSS
a.	/rana/	rná:	i-k-raná:	‘creek (NEG)’
b.	/ritisi/	ma-rtísi	i-k-ritsí:	‘to shock (NEG)’
d.	/giŋigiŋi/	gŋígŋi	i-k-giŋigiŋí:	‘longan (NEG)’
e.	/takasuluŋu/	tkaslúŋu	i-k-taksulŋú:	‘shrimp (NEG)’

Maga affixes can broadly be categorized as ‘alternating’ or ‘non-alternating’, as summarized in Table (8). With non-alternating affixes, roots have the same surface pattern found in isolation stems, where the medial odd-numbered vowels of the root have deleted; for example, in the word [ma-rkími], the root is of the shape C₁_CV₂CV₃, implying that the first vowel of the root was deleted. In contrast, with alternating affixes, it’s the medial even-numbered vowels of the root that delete. For example, given [i-θilbí:], the root is of the shape CV₁C₂_CV₃. Hsin (2000) lists three alternating affixes: /i-/ ‘negation’, /l-/ ‘plural’, and /-a/ ‘imperative’. Verb-marking prefixes like /ma-/ and /u-/ are non-alternating, as are a few other less productive verbalizing prefixes.

(8) *Examples of alternating vs. non-alternating affixes*

Type	Affix	Affix gloss	Example	Gloss
Non-alternating	ma-	stative vb.	ma-rkími	‘surprised’
	u-	active vb.	u-θilbí	‘to fly’
	si-	to put on	si-kpíŋi	‘to put on clothes’
Alternating	i- -V	negative	i-θilbí:	‘NEG-to fly’
	-a:	imperative	sukl-á:	‘Pick up! (imperative)’
	l-	plural	l-válki	‘children’

The distinction between alternating and non-alternating affixes could be a consequence of the order in which they apply, rather than a fundamental difference between the two classes of affixes. In words that take multiple affixes, non-alternating affixes always appear closer to the root. Additionally, alternating affixes always attach to a root that already has a non-alternating prefix. Examples of attested affix/stem combinations are given in (9). If the domain of syncope is always the base to which the outermost affix attaches to, then for non-alternating prefixes, the domain of syncope will be identical to that of a bare stem. In contrast, the domain of syncope for alternating prefixes will always be different from that of a bare stem.

(9) *ATTESTED WORD COMBINATIONS*

Type	UR	Syncope domain	SR	gloss
<u>Bare root</u>	/siridiŋi/	(siridiŋi)	[srídŋi]	‘cradle’
non-alt + <u>root</u>	/si-kipiŋi/	si-(kipiŋi)	[si-kpíŋi]	‘put.on-clothes’
alt + non-alt + <u>root</u>	/i-si-kipiŋi/	i-(si-kipiŋi)	[i-s-kipŋí:]	‘NEG-put.on-clothes’

Maga rhythmic syncope alternations are likely to have developed from severe phonetic reduction of unstressed vowels. Similar pathways of diachronic change have been reported from languages like Nishnaabemwin (Valentine 2001, Bowers 2012). The picture that emerges is that at a point earlier in history, Maga Rukai had paradigms with no $\emptyset \sim V$ alternations, and therefore did not require positing abstract URs. The subsequent loss of metrically weak vowels then resulted in a synchronic pattern which requires composite URs.

2.3 Blocking of syncope

There are several regular (phonologically-motivated) exceptions to syncope. First, syncope is blocked (near-exceptionlessly) in onsetless syllables. Examples of this are given in (10) for both word-initial vowels (10a), and medial vowels (10b). Vowels that otherwise would have been deleted (i.e. is underlyingly in an odd-numbered syllable) are underlined, and the expected form, had syncope applied, is given in the column titled ‘Expected SR’. Following Chen (2008), this can be characterized as the effects of a positional faithfulness constraint MAX-V/[σ], which penalizes deletion of syllable-initial vowels.

There is one systematic exception to this restriction against deletion of onsetless vowels; as discussed above in Section 2.2, in negative forms of active verbs, the active verb prefix /u-/ deletes, as in /i-u-ciŋili-V/ → [i-ciŋlɛ:] ‘to not see’.

(10) Onsetless vowels do not delete

a. Initial vowels (N = 120/124; 97%)

UR	EXPCTED SR	OBSERVED SR	GLOSS
/abu/	<u>_</u> bú	<u>á</u> bu	‘ash’
/uvaci/	<u>_</u> váci	<u>u</u> váci	‘vein’

b. Medial onsetless syllables (N = 37/37; 100%)

UR	EXPECTED SR	OBSERVED SR	GLOSS
/liθieki/	lθi_ki	lθi <u>ie</u> ki	‘waterfall’
/i-ka-nuəŋi/	i-k-nu_ŋɛ:	i-k-nu <u>ə</u> ŋɛ:	‘NEG-cattle’

Vowels in hiatus also do not delete; examples are given in (11). First, high vowels followed by an /a/ tend to become glides; examples of glide formation are given in (11a). Glide deletion is variable; Hsin’s data contains many forms transcribed with high vowels where glides are expected, such as [u-piárci] ‘shout’, and [buésə] ‘rice’. Other vowel-vowel sequences surface faithfully, even when one of the vowels is in a deleting position, as shown in (11b)

(11) *Vowels in hiatus do not delete*

a. *Glide formation of {i,u}a sequences (N = 35/54, 68%)*

UR	EXPECTED SR	OBSERVED SR	GLOSS
/u-puara/	u-p_árá	u-pwára	‘to capture’
/u-θiabi/	u-θ_ábi	u-θyábi	‘to chop’

b. *Hiatus in other VV sequences (N = 33/35, 94%)*

UR	EXPECTED SR	OBSERVED SR	GLOSS
/nuəŋi/	n_éŋi	nuéŋi	‘cattle, cow’
/kiudu/	k_údu	kiúdu	‘ride (a vehicle)’

Additionally, geminates are dispreferred in Maga, though attested in a few monomorphemic words (e.g. [búttu] ‘squirrel’, [tómma] ‘dry field’, [brúccu] ‘egg’). There is a resulting effect of antigemination (McCarthy 1986), where syncope is blocked if it would result in a geminate consonant; examples of this are given in (12). There is also a contrast between initial and medial geminates; syncope is always blocked if it would result in an initial geminate (12a), but only variably blocked in the case of medial geminates (12b). The difference in initial and medial geminates can be attributed to a typologically common and phonetically-motivated dispreference for initial geminate consonants (Muller 2001).

(12) *Deletion is blocked by geminates*

a. *Initial geminates (N = 25/25; 100%)*

UR	EXPECTED SR	OBSERVED SR	GLOSS
/tutuku/	t_túku	tutúku	‘rabbit’
/rarami/	r_rámi	rarámi	‘bird’

b. *Medial geminates in stems (N = 61/83; 73%)*

UR	EXPECTED SR	OBSERVED SR	GLOSS
/θelevava/	θlév_va	θleváva	‘rainbow’
/valiruru/	vlír_ru	vlirúru	‘glutinous rice’

2.4 Vowel lowering

There is an additional morphophonemic alternation that interacts with rhythmic syncope, but is not of direct interest to the current paper, where high vowels /i, u, i/ lower to mid vowels [e, o, ə]. Examples of vowel lowering are given in (13). Following Hsin’s 2003 description, all occurrences of /i, u, i/ following an /a/ are lowered to [e, o, ə] respectively, but *only* if the /a/ is deleted. In words like (13a-b), this causes lowering of /i/ in the stem form, while in words like (13c), lowering happens in the negative form. Words like [amíci] (*[améce]) ‘tree root’ and [págu] (*[págo]) ‘gallbladder’ demonstrate that lowering does not happen if the preceding /a/ is not deleted.

(13) *Vowel lowering (alternating vowels are underlined)*

	UR	STEM	NEGATIVE	GLOSS
a.	/damili/	dm <u>é</u> le	i-k-daml <u>í</u>	‘hemp (NEG)’
b.	/valu/	vl <u>ó</u>	i-k-val <u>ú</u>	‘bee (NEG)’
c.	/alapi/	alá <u>p</u> i	i-k-al <u>pé</u>	‘slate (NEG)’

3 Learnability of rhythmic syncope in Maga

Rhythmic syncope is potentially challenging to learn for several reasons. Rhythmic syncope is formally complex (Bowers & Hao 2020) and often opaque, as syncope must happen after building the metrical structure which conditions it (Kager 1997, McCarthy 2008). This is challenging to account for in parallel theories of phonology like OT. Syncope can be transparent and generable in parallel frameworks if it is driven by constraints (or rules) that are surface-satisfied; see for example analyses of Southeastern Tepehuán (Kager 1997, Gouskova 2003). However, constraints on surface markedness cannot motivate syncope in Maga.

First, Chen (2008) shows that segmental markedness constraints, such as avoidance of consonant clusters, cannot motivate rhythmic syncope in Maga. Constraints on surface metrical structure also do not straightforwardly account for the syncope pattern, because the location of surface stress does not always match the location of syncope. Maga syncope is assigned on odd syllables counting from the left, but stress is counted from the right edge, surfacing on the penultimate mora. As a result, for words of even syllable parity, syncope is inconsistent with surface stress. This inconsistency between stress and syncope is schematized in (14). Under the UR column, the position that should have been stressed (i.e. is in penultimate position) is underlined. For stems that have underlying even parity, syncope targets the position that should have been stressed, had deletion not occurred.

This is a type of *metrical incoherence*, where a metrically conditioned process (in this case syncope) does not appear to refer to surface stress (Dresher & Lahiri 1991, Gordon 2016, Kaplan 2022). As a result, there is no straightforward surface-oriented solution that can derive both stress and syncope. Prior analyses of Maga syncope have therefore opted for solutions that are derivational (Hsin 2000) or do not rely on purely phonological constraints (Chen 2008).

(14) *Interaction of syncope and stress assignment by syllable parity*

Parity	UR	SR	Example		
Even	/σ ₁ σ ₂ /	[∅ ₁ 'σ ₂]	/t í ba/	tb á	‘wild yam’
	/σ ₁ σ ₂ <u>σ₃</u> σ ₄ /	[∅ ₁ 'σ ₂ ∅ ₃ σ ₄]	/tavan <u>u</u> lu/	tv án lu	‘chili pepper’
Odd	/σ ₁ <u>σ₂</u> σ ₃ /	[∅ ₁ 'σ ₂ σ ₃]	/d am ari/	d má ri	‘moon’
	/σ ₁ σ ₂ σ ₃ <u>σ₄</u> σ ₅ /	[∅ ₁ σ ₂ ∅ ₃ 'σ ₄ σ ₅]	/takasul <u>u</u> du/	tkasl ú du	‘shrimp’

The opacity presented by rhythmic syncope has been argued to make it less learnable, and therefore diachronically unstable. For example, Bowers (2012, 2019) attributes the breakdown of rhythmic syncope in Nishnaabemwin to the opacity of the pattern. Similarly, Kaplan (2022) finds that in Southern Pomo, a subset of paradigms has been reanalyzed, turning a metrically incoherent syncope pattern into one that is motivated by constraints on surface metrical structure.

Finally, the morpheme-specific nature of Maga rhythmic syncope may present additional challenges to its learnability. On the surface, some affixes appear to trigger vowel alternations, while others don't. Additionally, even within the set of alternating affixes, deletion doesn't always happen on odd-numbered vowels. In Hsin's analysis, these facts are explained by positing that the domain of syncope is not the word, but rather the base to which the outermost affix attaches to; as a result, the surface deletion pattern differs for monomorphemes and complex words. Affixes also differ from roots in other respects; the active prefix /u-/ deletes in syncope positions, violating a general restriction against onsetless vowel deletion. In other words, speakers must learn that syncope is morphologically restricted, and applies differently to roots and affixes.

These factors, taken together, may obscure the surface syncope pattern, making it more challenging for learners to acquire. As a result, the pattern may be prone to restructuring; in the present study, I look at these patterns of restructuring and find that restructuring systematically occurs in a way which results in more concrete URs.

4 Formalizing degrees of UR abstraction in Maga

To formalize degrees of abstraction, I adopt Kenstowicz and Kisseberth's (1977: ch. 1) taxonomy of UR abstractness, which defines levels of abstractness in terms of how divergent a UR is from its associated SRs; following Wang & Hayes (2025), I refer to this hierarchy as the KK-hierarchy. The KK-hierarchy not only allows for a finer-grained evaluation of how abstract URs can be in Maga, but also allows for comparison with recent work by Wang & Hayes (2025), who propose a model of UR learning that is situated in the same hierarchy. In this section, I give an overview of the KK-hierarchy, then and discuss possible outcomes of restructuring for Maga under each level of this hierarchy.

4.1 The KK-hierarchy

As a baseline, consider the following two paradigms, representing the input that Maga speakers received following the development of rhythmic syncope. Boldface vowels are only present in the stem form, while outlined vowels are only present in the negative. Given such inputs, speakers then infer URs; depending on the level of abstractness allowed in UR learning, these paradigms could be restructured, resulting in the loss of formerly contrastive information.

- (15) *Input to learners (observed forms)*
- | STEM | NEGATIVE | GLOSS |
|-------|--------------------------|--------------|
| ɖmari | i-k-ɖa [̣] mri: | ‘moon (NEG)’ |
| tmusu | i-k-ti [̣] msu: | ‘salt (NEG)’ |

KK-A: only invariant properties of allomorphs

Under KK-A, the UR of a morpheme “consists of all and only the invariant phonetic properties of that morpheme’s various SRs” (Kenstowicz & Kisseberth 1977: p. 8). This is highly restrictive, disallowing even standard phonemicization. For example, English “pan” [pæ̃n], where the vowel is predictably nasalized, would have the UR /pæ̃n/. KK-A’, a less restrictive version of KK-A, allows predictable allophony to be treated as unspecified.

Both KK-A and KK-A’, when applied to Maga Rukai, would not allow any of the alternating vowels to be present in the UR. A root like ‘moon’ would have the UR /ɖmri/, whose only vowel is the non-alternating stem-final vowel. Under a KK-A approach to UR learning, URs should be reanalyzed in a way that removes all contrastive information about the alternating vowels. This could result in a hypothetical grammar like (16), where formerly contrastive vowels have been replaced with a predictable epenthetic vowel like [i].

- (16) Hypothetical Grammar A: retain non-alternating parts of the paradigm (grey vowel is predictable)
- | UR-A | | STEM | NEGATIVE |
|--------|---|-----------------------|----------------------------|
| /ɖmri/ | → | [ɖmi [̣] ri] | [i-k-ɖmi [̣] ri:] |
| /tmsu/ | → | [tmi [̣] su] | [i-k-ti [̣] msu:] |

KK-B: single allomorph

Broadly, KK-B requires that URs be based on the same surface allomorph (i.e. the same slot in a paradigm). Under the basic version of KK-B, the UR must be based on the isolation stem, or as close to the isolation stem as the language permits (Kenstowicz & Kisseberth 1977: p. 11). The less restrictive KK-B’ states that the learner should base the UR on the allomorph that appears in the most contexts, while KK-B”, which has been extensively explored in work by Albright (2002a,b, 2010: et seq.), restricts URs to be based on the most informative slot in a paradigm.

The subtypes of KK-B, when applied to Maga, predict that URs either only retain vowels present in the stem form, or vowels present in the negative form, but not both. Grammar B1, for example, shows a scenario where URs correspond to what was formerly the stem allomorph, and retain only vowels that were present in the stem allomorph (shown in bold). Paradigms have leveled towards the stem form, meaning that in both the stem and the negative forms, the root surfaces as was formerly the stem allomorph. In Grammar B2, paradigms have instead leveled towards

the negative form, and URs retain only vowels that were present in the negative (shown in box outlines). Note that while grammars B1 and B2 have leveled out vowel alternations, this is not a requirement of KK-B. KK-B also allows for grammars like B3, where $\emptyset \sim V$ alternations remain, but are completely predictable from one surface allomorph; in B3, the UR is based on the former stem allomorph, and vowels that surface in the negative form are copies of the stem vowels.

(17) Hypothetical KK-B grammars

- a. Grammar B1: leveling towards the stem allomorph.

UR	STEM	NEGATIVE
/ɖmari/	→ [ɖmári]	[i-k-ɖmarí:]
/tmusu/	→ [tmúsu]	[i-k-tmusú:]

- b. Grammar B2: leveling towards negative allomorph.

UR	STEM	NEGATIVE
/ɖámri/	→ [ɖámri]	[i-k-ɖámrí:]
/tímsu/	→ [tímsu]	[i-k-tímsú:]

- c. Grammar B3: $\emptyset \sim V$ alternations are predictable from the stem allomorph

UR	STEM	NEGATIVE
/ɖmari/	→ [ɖmári]	[i-k-ɖamrí:]
/tmusu/	→ [tmúsu]	[i-k-tumsú:]

KK-C: choosing among allomorphs

In the KK-C level of the hierarchy, the UR of a morpheme is always based on some surface allomorph, but this allomorph does not need to come from the same paradigm slot for every morpheme. For Maga, this allows for grammars like (18), where some words have leveled towards the stem allomorph while others have leveled towards the negative allomorph.

- (18) Hypothetical Grammar C: leveling towards either the stem or negative allomorph.

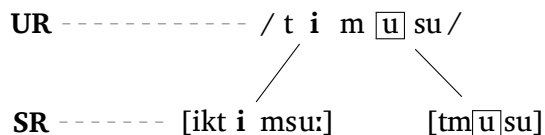
UR	STEM	NEGATIVE	
/ɖmari/	→ [ɖmári]	[i-k-ɖmarí:]	(base = stem)
/tímsu/	→ [tímsu]	[i-k-tímsú:]	(base = negative)

KK-D: segmentally composite URs

At KK-D, URs must be composed of segments that surface in some allomorph, but not necessarily the same one. This is the level required for a traditional analysis of the Maga syncope pattern. As schematized in (19), for words like ‘salt’, no single surface allomorph has information about all the underlying vowels. As a result, the UR must take V1 from the negative allomorph, and V2

from the stem allomorph. If the KK-D level of the hierarchy is learnable, then we would predict Maga syncope to be stable over time.

(19) *Composite UR of ‘salt’*



KK-E/F: featurally composite URs

Under KK-E/F of the abstractness hierarchy, underlying forms can have abstract phonemes which never occur in the surface allomorphs. This level of the hierarchy does not directly concern the current paper, since the Maga pattern does not require a higher level of abstraction than KK-D.

Multiple listed URs (KK-C')

The KK hierarchy assumes that learners have just one UR corresponding to each morpheme. However, it is also possible that learners list multiple underlying representations for a given morpheme, each corresponding to a surface allomorph (e.g. Zuraw 2000, 2010, Moore-Cantwell & Pater 2016, Kuo 2023b). For example, the paradigm [dmáři]~[i-k-d[amrí:] may be associated with two listed allomorphs, /dmari/ and /d[amri/. In terms of learning difficulty, a listed allomorph approach would be similar to (or potentially easier than) a single-allomorph approach (i.e. KK-B), since learners can simply list any allomorph they encounter, without considering other alternatives. However, this approach would predict restructuring that is more similar to the changes predicted under KK-C. That is, we may expect to see leveling towards both the stem and negative allomorphs, since both allomorphs are listed in the grammar.

4.2 Degree of abstraction required in the Maga lexicon

Notably, although many paradigms in Maga require positing segmentally composite URs (KK-E), some paradigms are still predictable from one or both surface allomorphs. Table (20) provides examples of paradigms that vary in their degree of abstractness.

First, roots that are underlyingly /V1 CV2/ can be accounted for using KK-A URs. As shown in (20a), these paradigms have no vowel alternations; V1 in the stem does not delete because it's onsetless, while in the negative form, neither vowels are in positions for syncope. Roots of the shape /CV1 CV2/ (20b) never undergo syncope in the negative allomorph, so are fully predictable from this allomorph. Roots of the shape /V1 CV2 CV3/ (20c) are fully predictable from the stem allomorph; V1 does not delete because it is onsetless, and the other vowels are not in deleting positions. Blocking of syncope from vowel hiatus and/or antigemination can also result in surface allomorphs that contain information about all the underlying vowels.

- (20) *Summary of UR types required in the lexicon, assuming regular application of syncope (A: invariant only; B:single surface allomorph; C: either surface allomorph; D: segmentally composite)*

	Predictable	KK						
	allomorph	level	UR	stem	negative	gloss	N	P
a.	both	A	/abu/	ábu	i-abú:	‘ash (NEG)’	84	8%
b.	negative	B/C	/cimi/	cmí:	i-k-cimí:	‘cheek (NEG)’	96	9%
c.	stem	B/C	/amici/	amíci	i-k-amcí:	‘root (NEG)’	230	22%
d.	neither	D	/dukaci/	dkáci	i-k-dukcí:	‘mud (NEG)’	639	61%

Table (20) additionally summarizes the degree of abstractness required in the Maga corpus, if syncope applies regularly (i.e. no restructuring has occurred). If Hsin’s corpus is taken to reflect the general distributions of the Maga lexicon, then the majority of the lexicon requires positing abstract URs of some kind. Only 8% of the corpus can be accounted for under KK-A. In contrast, over 60% of words can only be accounted for under KK-D of the hierarchy.

5 Restructuring of rhythmic syncope alternations

Assuming regular application of syncope, over half of the Maga lexicon would require composite URs under a traditional morphophonemic analysis. However, it turns out that for the subset of forms where both the stem and negative allomorphs are known, about 24% (190/790) of stem forms and 23% (185/790) of negative forms fail to undergo the regular syncope pattern. Overall, 40% (314/790) of paradigms have at least one member that does not undergo syncope regularly. This irregularity suggests that substantial restructuring has occurred. In fact, these numbers do not reflect the full extent of restructuring in Maga, as some paradigms have also maintained what looks on the surface to be a regular rhythmic syncope alternation, but undergone changes in vowel-quality.

In this section, I describe several types of restructuring observed in Maga, and argue they all result in paradigms which require less abstract URs. First, as discussed in Section 5.1, some paradigms have leveled towards either the stem or negative allomorph, removing $\emptyset \sim V$ alternations. Additionally, in Section 5.3-5.5, I show that speakers have extended a ‘vowel-matching’ alternation which makes paradigms more concrete.

The data I use is the same corpus of 1049 words above; for some of these words, the negative allomorph is not available. In total, a subset of 790 words are listed with the stem/negative paradigm; for these, we can infer the direction of restructuring. In the rest of this section, I will also make a distinction between the ‘idealized UR’ and ‘new UR’. Idealized URs are those proposed under a traditional morphophonemic analysis, assuming regularly alternating paradigms; these will be represented using double slashes. New URs are the more concrete URs proposed to have been acquired by are Maga speakers, following restructuring of paradigms. Given a paradigm like [ɖmári]~[i-k-ɖamrí:], for example, the idealized UR is //ɖamari//, while a possible new UR is

/ɬmari/.

5.1 Paradigm leveling

As previewed above, many paradigms have leveled out vowel alternations. Direction of leveling is inferred from the syncope pattern found in Hsin’s corpus. For example, given the word ‘salt’ in (21), there are three possible syncope patterns (not leveled, leveled to stem, leveled to negative) given in (21); in these examples, the leveled-out form is shaded in grey. As with before, vowels present only in the stem form (before restructuring/leveling) are shown in bold, while vowels present only in the negative are outlined. In (21b), the negative allomorph does not show the regular syncope pattern, and instead is identical to the stem allomorph; additionally, the paradigm only retains vowels of the (former) stem allomorph. Taken together, this implies that leveling towards the stem has occurred. Example (21c) shows leveling in the opposite direction. Leveling always results in a more concrete paradigm, as it removes all vowel alternations.

(21) *Hypothetical types of leveling for ‘salt’* //tɪmusu//

PATTERN	PARADIGM	NEW UR
a. Not leveled (regular syncope)	[t m úsu] ~ [i-k-t <u>i</u> msú:]	–
b. Leveled to stem	[t m úsu] ~ [i-k-t m usú:]	/tmusu/
c. Leveled to negative	[t <u>i</u> msu] ~ [i-k-t <u>i</u> msú:]	/timsu/

Table (22) summarizes the proportion of forms that have been leveled; the row titled ‘other’ shows cases which have irregular syncope but not paradigm leveling; these are discussed in Section 5.2. Paradigms are organized by the type of idealized UR they would have prior to leveling. For example, (22b) represents a case where the underlying vowels are predictable from the stem allomorph, and leveling towards the stem has occurred. The ‘regular’ negative form should be [i-udlǫ́:], but instead, it has leveled towards the stem and is now [i-udalǫ́:]. On the other hand, ‘quench’ in (22c) is also predictable from the stem, but has leveled towards the negative allomorph, such that the isolation stem is [ǫ́ɬpi], with the syncope pattern of the negative form.

Looking at Table (22), as expected, non-alternating forms have not undergone any changes. Negative-base paradigms (i.e. ones predictable from the negative allomorph) almost always leveled towards the negative allomorph, if they were leveled (22f). Similarly, stem-base paradigms almost always leveled towards the stem (22b). For the composite UR forms, there has generally been less leveling (N=63, 14%).³ Where leveling has occurred, it has happened towards both allomorphs, with a bias towards the stem allomorphs.

³Note that this is also the subset of forms that underwent vowel-matching restructuring.

- (22) **Patterns of leveling.** The ‘Old’ column is what paradigms should have looked like with regular syncope; ‘New’ shows the currently attested paradigm. Leveled out forms are shaded grey.

Ideal UR	Leveled?	N	P	Example		
				Old	New	gloss (stem)
a. non-alt.	no	61	100%	N/A	ábu~i-abú:	‘ash’
b. stem-base	to stem	33	39%	událi~i-udlí:	událi~i-udalí:	‘rain’
c.	to neg.	1	1%	id̥pi~i-idpí:	id̥pi~i-idpí:	‘quench’
d.	no	51	60%	N/A	amíci~i-k-amcí:	‘root’
e. negative-base	to stem	1	1%	u-kcyá:~i-kūcyá:	u-kcyá:~i-kcyá:	‘cut’
f.	to neg.	70	43%	u-ktí:~i-kī́tí:	u-kī́tí~i-kī́tí:	‘slice’
g.	no	93	56%	N/A	bró:~i-k-bāró:	‘lake’
h. composite	to stem	36	8%	u-rbáŋi~i-rābŋí:	u-rbáŋi~i-rbaní:	‘trap’
i.	to neg.	27	6%	u-θl̥íbi~i-θl̥ílbí:	u-θl̥íbi~i-θl̥ílbí:	‘fly’
j.	other	96	20%	ma-lŋílo~i-k-l̥íŋló:	ma-lŋílo~i-k-l̥íŋíló:	‘thin’
k.	no	319	67%	N/A	q̥mari~i-k-q̥a mri:	‘moon’

Overall, there has been extensive leveling, resulting in more concrete paradigms. In terms of the direction of leveling, stem-base paradigms tend to level towards the stem allomorph (N = 33/34 leveled items), and negative-base paradigms tend to level towards the negative (N = 70/71). For composite UR forms that are predictable from neither allomorph, there is a slight preference for leveling towards the stem. Since leveling has happened in both directions, results are compatible with either a KK-C (choosing among allomorphs) or listed allomorphs view of UR selection.

Interestingly, composite UR forms have leveled less than other forms. If restructuring is driven by abstractness-avoidance, we should expect exactly these forms to be restructured the most. In Sections 5.3-5.5, I propose that composite UR paradigms have been restructured in other ways, which preserve the surface vowel alternation patterns but still result in a more concrete analysis.

5.2 Partial leveling

Some paradigms in Hsin’s corpus have irregular syncope, but do not appear to have leveled towards either allomorph; these are labeled as ‘other’ in Table (22) above. Of these 96 cases of irregular syncope, most (N = 87/96) occur with longer words (where the ideal UR of the root has four or more syllables). With these longer words, the majority of cases (N = 81/90) look like ‘partial leveling’, where the paradigm has preserved vowel alternations at the left edge of the root, but the negative allomorph has otherwise leveled towards the stem form. Representative examples are given in (23); in these examples, the stem is regularly alternating, while the negative allomorph has been partially leveled; under ‘Negative allomorph’, the ‘Regular’ column shows the expected negative allomorph, if syncope had regularly applied. ‘Observed’ shows the actual observed form; comparison of these two columns shows that the partially leveled negative allomorphs differ from the stem only in that a vowel is inserted between the first two consonants.

- (23) **Examples of partial leveling.** In all cases, the stem has the regular syncope pattern. In the negative allomorph, vowels that are not in the stem allomorph are outlined in boxes.

Stem	Negative allomorph			Gloss (stem)
	Regular	Observed	Unattested	
ma-θlímdi	i-k-θílmídí:	i-k-θílimdí:	i-k-θlimídí:	‘tasty’
u-grígrí	i-gírgírí:	i-gírigrí:	i-grigírí:	‘cut, saw’
spégsi	i-k-sápgísí:	i-k-sápegsí:	i-k-spegísí:	‘neighbor’
θtíθpí	i-k-θátθípí:	i-k-θátíθpí:	i-k-θtθípí:	‘index finger’

Partial leveling does not necessarily result in more concrete URs, as alternations remain between the stem and negative members of the paradigm. However, it turns out that the majority of partial leveling cases (N=85/96) are amenable to a concrete-UR analysis, as will be discussed in Section 5.3. More generally, the partial leveling cases are also informative for understanding how learners have potentially restructured Maga vowel syncope alternations. In particular, in all partial leveling cases, paradigms have maintained vowel alternation at the *left* edge of the root. Table (23) also shows, for each partially leveled paradigm, an example of an unattested outcome of restructuring, which maintains alternations at the *right* edge of the root but levels out other alternations. This provides support for the idea that speakers have learned a left-edge alternation, rather than a general rhythmic syncope alternation. This idea is fleshed out in detail in Section 6.

It’s plausible for learners to have acquired a left-edge alternation, because the majority of roots can be analyzed this way. Shorter roots (meaning 2-3 syllables in the idealized UR) make up around 70% of the Maga lexicon. For these shorter roots, rhythmic syncope looks exactly the same as a left-edge alternation. For example, given a paradigm like [u-çŋíli]~[i-ciŋlí:], learners could either derive the surface forms via rhythmic syncope, or via a generalization like “insert a vowel between the first two consonants of the negative form.” In sum, a large chunk of the lexicon was already compatible with a left-edge alternation analysis.

5.3 Vowel-matching and surface predictability from the stem form

For a subset of paradigms, the surface alternation pattern is compatible with alternate analyses that remove the need for proposing composite URs. In addition to the leveling described in the previous section, speakers also appear to have extended these alternate concrete analyses. Note that in the following descriptions, I assume that the base of derivations is the stem allomorph, meaning that these alternate analyses make alternations *predictable from the stem*. Similar concrete sub-patterns can be found if the negative allomorph is assumed to be the base of derivation, but I focus on stem-based analyses for two reasons: i) results from the previous section suggest that there is a slight preference for leveling towards the stem allomorph, and ii) as will be described in Section 5.4, the direction of restructuring for these alternate analyses is overwhelmingly towards the stem allomorph.

First, in stem-base paradigms, the negative form can straightforwardly be derived from the stem

form, which preserves all vowel contrasts. For these paradigms, the surface pattern resembles deletion; given a word like //abaki// [abáki~i-k-ab_kí:], the negative form can be derived from the stem by simply deleting the second vowel.

Another notable pattern in the Maga lexicon is that the leftmost vowel of surface allomorphs within a paradigm tend to match each other; examples of this are given in (24), where the relevant vowels are bolded. In (24a), one vowel surfaces only in the stem, while another surfaces only in the negative, but both of these vowels are [i]. In (24b), the first vowel of the root only surfaces in the negative form, but is identical to the second vowel (which surfaces in both allomorphs). In Hsin's corpus, 388/723 paradigms with vowel alternations are vowel-matching. This tendency is stronger for paradigms with composite ideal URs (60%, n = 244/400).

(24) *Vowel matching examples*

	PARADIGM TYPE	IDEAL UR	STEM	NEGATIVE	GLOSS (STEM)
a.	composite	//cɪŋ i li//	u-cŋ i li	i-cɪŋ i lí:	'see'
b.	negative-base	//r a na//	rn a :	i-k-r a ná:	'creek'

Crucially, if vowel-matching is a strong enough tendency, it would render a subset of paradigms predictable from just the surface allomorphs, removing the need for abstract URs. First, in negative base paradigms that are vowel-matching, the surface pattern can be described as vowel-copying; as shown in (25), if the URs for these paradigms are assumed match the stem form (in this case /rna/), the negative form can be derived by copying the first vowel. Note that the derivations in (25) are very general; specific constraints motivating vowel deletion in verb markers, and copying and metathesis, will be introduced in Section 6. These paradigms are already predictable from the negative allomorph, but vowel-matching makes them additionally predictable from the stem.

Next, for paradigms that should require a composite UR, vowel-matching causes the surface pattern to look like metathesis. For example, if we assume that the UR of 'see' is /cŋili/ (corresponding to the stem allomorph), the negative form can be derived by metathesis of [i] and [ŋ]. If vowel-matching is predictable, then these paradigms become predictable from just the surface allomorphs, removing the need for abstract URs.

(25) *Deriving negative forms in a concrete analysis of vowel-matching paradigms*

Negative-base	UR: //rana// → /rna:/ (= stem allomorph)
	Derivation: /i-ka-rna ₁ -V/ → [i-k-r a ₁ n a ₁]
Composite-UR	UR: //cɪŋili// → /cŋili/ (= stem allomorph)
	Derivation: /i-u-cŋ ₁ i ₂ li-V/ → [i-ci ₂ ŋ ₁ li:]

The majority of partially leveled paradigms (Section 5.2) can also be analyzed as vowel copying, making them concrete. An example derivation is given in (26); (26a) shows the expected surface paradigm and composite UR. In the actual observed form (26b), the negative allomorph

differs from the stem by just one vowel; in fact, if we assume that the UR corresponds to the stem allomorph, the negative form can be derived by copying the first underlying vowel.

(26)

	stem	negative	UR
a. Expected	[r ₁ ma ₂ r ₂ mo ₄]	[i-k-ra ₁ m ₂ ra ₃ mo ₄]	//ra ₁ ma ₂ ra ₃ mo ₄ //
b. Partially leveled	[rma ₁ rmo ₂]	[i-k-r _{a₁} m _{a₁} rmo ₂]	/rma ₁ rmo ₂ /

Table (27) summarizes the concrete sub-patterns found in Maga; counts are taken from Hsin's corpus, *assuming regular application of syncope*. Over half of the lexicon (63%) is predictable from these concrete sub-patterns. Slightly under half of the paradigms (N = 307, 43%) can be analyzed as undergoing metathesis, while deletion and copying explain a smaller proportion of the data.

(27) *Alternate analyses of Maga paradigms. Counts reflect the number of forms in Hsin's corpus that fit each analysis, if syncope is assumed to apply regularly*

	BASE	SURFACE PATTERN	N	P	EXAMPLE
a.	Stem	Deletion	84	10%	abáki~i-k-ab_kí:
b.	Negative	Copying	70	12%	rná:~i-k-raná:
c.	Neither	Metathesis	301	41%	rnígi ~ i-k-ríngí:
d.	(non-matching)	None	271	37%	tbusu~i-k-tibsu: sro:~i-k-suro:

If speakers do in fact prefer to learn concrete URs, even if this results in an imperfect grammar with exceptions, then we might expect these concrete analyses to be extended throughout the lexicon. To preview the results of Section 5.4-5.5, restructuring has in fact caused the two patterns that rely on vowel-matching, metathesis and copying, to be extended.

5.4 Direction of vowel-matching changes

This section compares Proto-Rukai forms with their Maga Rukai reflexes, to first confirm that vowel-related restructuring has generally resulted in more vowel-matching. Additionally, I show that the direction of vowel-matching changes is overwhelmingly towards the stem allomorph, suggesting a bias for learning stem-based URs.

5.4.1 Data

To infer the direction of vowel changes over time, I compare Proto-Rukai forms with their known Maga reflexes. Mismatches in the vowels of a protoform and its reflex indicate that vowel-related restructuring has occurred. Importantly, no other dialect of Rukai has developed vowel syncope alternations. Tona, the dialect most closely related to Maga, has undergone no vowel-related

changes. As a result, Proto-Rukai can be reliably used to infer what Maga vowels looked like prior to any restructuring that came after the development of rhythmic syncope.

As with the above sections, the Maga data comes from Hsin (2000). Data on Proto-Rukai comes from Li's 1977 list of 532 protoforms, reconstructed from five of the six Rukai dialects⁴. Table (28) lists Proto-Rukai vowels and their regular reflexes in Maga; in the rest of the paper, for comparability with the Maga data, Proto-Rukai vowels are presented and organized by their reflexes in Maga, as summarized in (28).

(28) *Vowel correspondences, Proto-Rukai and Maga*

PROTO-RUKAI	MAGA	EXAMPLE
*a	a	*kav <u>a</u> > kvá: 'hoe'
*i	i	*rigi > rgí: 'horse'
*o, oa	u	*tim <u>oso</u> > tmús <u>u</u> 'salt'; * <u>o</u> alopo > ulúpu 'hunt'
*ə	ɪ	*pələŋə > plínɪ 'ghost'
*(a...)i	e	*caki > cké: 'excrement'
*(a...)o	o	*val <u>o</u> > vlú 'bee'
*(a...)ə	ə	*baləbalə > bláblə 'bamboo'
*ay, *ai	e, e:	*acil <u>ay</u> > acíle 'water'; *pag <u>ay</u> > pgé: 'rice plant'
*aw, *ao	o, o:	*maram <u>aw</u> > marámo 'same'; *ko <u>aw</u> > kuó: 'eagle'

In general, there is a straightforward correspondence between Proto-Rukai and Maga, with the exception that mid vowels *o and *ə are reflected in Maga as the high vowels [u] and [ɪ]. Proto-Rukai diphthongs like *aw and *ay are reflected as monophthongs in Maga, variably as short or long vowels. The one wrinkle is that *i, *o, *ə are reflected as mid vowels [e], [o], and [ə] if they follow a *deleted* low vowel (*a)⁵. These mid vowel reflexes are generally not observed if preceding [a] has not been deleted (e.g. *aŋato > aŋatu 'tree'), though there are a few exceptions where *o and *i adjacent to un-deleted *a are reflected as mid vowels (e.g. *aθo > aθó: 'dog').

5.4.2 Results

Comparison of Proto-Rukai forms and their Maga reflexes first confirms that where vowel-related restructuring has occurred, it has primarily resulted in more vowel-matching, as opposed to some other type of change. The examples in (29) show the types of changes we are interested in, where the Maga paradigm does not show the expected vowel alternation, given the Proto-Rukai form. In these examples, the changed vowel is bolded and underlined.

⁴The Labuan dialect is not included in Li's wordlist

⁵See Section 2.4 for details on the resulting synchronic mid-vowel alternations

(29) *Examples: vowel changes from Proto-Rukai to Maga*

Proto-Rukai	expected UR	actual UR	paradigm	gloss (stem)
*samito	// <u>s</u> amitu//	// <u>s</u> imitu//	smítu~i-k-símtú:	‘lips’
*sinaw	//sin <u>o</u> sino//	//sin <u>i</u> sino//	u-snísno:~i-sinsino:	‘wash (clothes)’
*bicoka	//b <u>i</u> cuka//	//b <u>u</u> cuka//	bcúka~i-k-b <u>u</u> cká:	‘stomach’

In total, there were 55 forms where Proto-Rukai and Maga Rukai vowels mismatched. Table (30) summarizes the types and frequencies of observed changes. The majority of changes (~70%) resulted in more vowel-matching, consistent with the predictions of the previous section. This was confirmed in a Chi-Square goodness-of-fit test, which found that significantly more changes resulted in vowel-matching, $\chi^2(2, N = 55) = 33.6, p < 0.001$. Of the remaining changes, most neither increase nor decrease the rate of vowel-matching; for example, Proto-Rukai **ma-riday* should correspond to Maga [ma-rdé:]~[i-k-rídé:], but the observed paradigm is instead [ma-rdé:]~[i-k-rudé:].

(30) *Summary of vowel changes*

CHANGE RESULTS IN...	N	P	EXAMPLE
MORE vowel-matching	38	70%	* <u>s</u> amito → / <u>s</u> imitu/ ‘lips’
LESS vowel-matching	5	9%	*v <u>i</u> li → /v <u>u</u> li/ ‘leech’
Neither	12	21%	*ma-r <u>i</u> day → /r <u>u</u> de:/ ‘fast’

For a few words (N=5), vowel-matching forms were changed to be *non*-matching. Of these, two can plausibly be explained as the result of speakers extending a vowel-lowering alternation where high vowels lower to [e, o] following a deleted /a/ (see Section 2.4). For example, Proto-Rukai **ma-pilay* ‘tired’ should correspond to Maga [ma-plé:]~[i-k-pilé:] (UR = /pile/), but instead we observe [ma-plé:]~[i-k-palí:] (UR = /pali/). This could be because speakers, when given the stem [ma-ple:], interpreted the final [e] as the result of /i/-lowering after a deleted /a/.

Vowel-related restructuring can be either towards the stem or negative allomorphs. Table (31) below summarizes the observed directions of vowel changes, with representative examples for each direction; in these examples, the restructured allomorph is highlighted, and vowels that do not have the expected quality given the Proto-Rukai form are underlined. For example, (31a) represents a case of restructuring based on the stem allomorph, as speakers are extending the vowel that is present in the stem allomorph to the negative allomorph. The opposite scenario, where speakers extend the vowel present in the negative allomorph to the stem allomorph, would result in [smátu]~[i-k-samtú:]. For a few paradigms, the direction of restructuring is unclear because both V1 and V2 have changed. When these are excluded, restructuring has overwhelmingly been based on the stem allomorph (48/50, P = 96%). In contrast, just two vowel-related changes were based on the negative allomorph. Both of these cases, shown in (32), turn out to be potentially problematic.

(31) *Directions of vowel-matching reanalysis*

	BASE	N (P)	PROTO-RUKAI	EXPECTED SRS	ACTUAL SRS	GLOSS (STEM)
a. Stem		48 (87%)	* <u>sam</u> ito	smitu~i-k-samtu:	smitu~ik- <u>sim</u> tu:	‘lip’
b. Negative		2 (4%)	* <u>sinaw</u>	u-snosno~i-sinsino:	<u>u-sni</u> sno~i-sinsino:	‘wash’
c. Unclear		5 (9%)	* <u>tolar</u> a	tlari~i-k-tulri	<u>tlura</u> ~i-k- <u>talra</u> :	‘throat’

The source for ‘borrow’ is based on just one cognate from Tona, and potentially unreliable. Second, the word for ‘wash’ may have been influenced by a process of CiCo-reduplication that is found in other dialects of Rukai, where CiCo sequences are reduplicated as CiCi-CiCo (Zeitoun 2007, 2024). The Maga word for ‘wash’ fits the phonological profile for CiCo-copying, so the shift of *o to [i] may have been due to influence from dialects that have CiCo-copying.

(32)	Proto-Rukai	Expected SR	Observed	Gloss (stem)
	kyas <u>o</u> damə [Tona]	ks <u>ú</u> dmi~i-kesdamí:	ks <u>é</u> dmi~i-kesdamí:	‘borrow’
	* <u>sinaw</u>	u-sn <u>ó</u> sno~i-sinsinó:	<u>u-sni</u> sno~i-sinsinó :	‘wash’

Overall, the tendency for stem-based structuring suggests that speakers are learning concrete URs which correspond to the stem allomorph. For example, the root for ‘lip’ in (31a) is expected to have a UR //samitu// based on comparison with Proto-Rukai, but the direction of restructuring suggests that speakers have learned the UR /smitu/, corresponding to the surface stem.

5.5 Vowel-matching as abstractness-avoidance

There are two reasons why we might observe vowel-matching changes. The first is that there is some kind of general vowel-harmony restriction on Maga vowels, which may have been present already in an earlier stage of Rukai. The second possibility is that speakers are extending vowel-matching specifically to remove abstract URs; in this case, we should expect to see an increase in vowel-matching only when it would increase a paradigm’s predictability from a surface allomorph. Because vowel-matching changes have almost been all based on the stem form (Section 5.4), they should specifically happen when they increase a paradigm’s predictability from the stem form.

As summarized in Table (33), in composite UR paradigms, vowel-matching allows the negative form to be derived from the stem form via metathesis. In negative-base paradigms, vowel-matching makes the negative form predictable from the stem via vowel-copying. In contrast, stem-base and non-alternating paradigms are already predictable from the stem, regardless of whether vowel-matching is present. If speakers are extending vowel-matching to remove abstract URs, rather than due to general vowel harmony pressures, vowel-matching should only increase in composite-UR and negative-base paradigms.

(33) *The effect of vowel-matching on a paradigm's predictability from the stem allomorph*

	TYPE	EFFECT	EXAMPLE
a.	Composite UR	More predictable (metathesis)	rnígi ~ i-k-ringí:
b.	Negative-base	More predictable (copying)	rná:~i-k-raná:
c.	Stem-base	no effect	ubúlu~i-k-ublú:
d.	Non-alternating	no effect	ába~i-k-abá:

To test this prediction, I compare the distribution of vowel-matching forms in Proto-Rukai and Maga. In my comparison, I focus on comparing the leftmost vowels of the stem and negative allomorphs (V1 and V2 of the ideal UR, e.g. //u/ and //a// in //dukaci// ‘mud’). This data, though slightly pared down, should still be a good reflection of vowel-matching patterns across Maga, since for most roots, these are the only vowels that alternate (70%, N = 735/1049). Additionally, the last two vowels of roots have a tendency to match due to a process of echo epenthesis that is not the focus of the current study (Li 1977b).

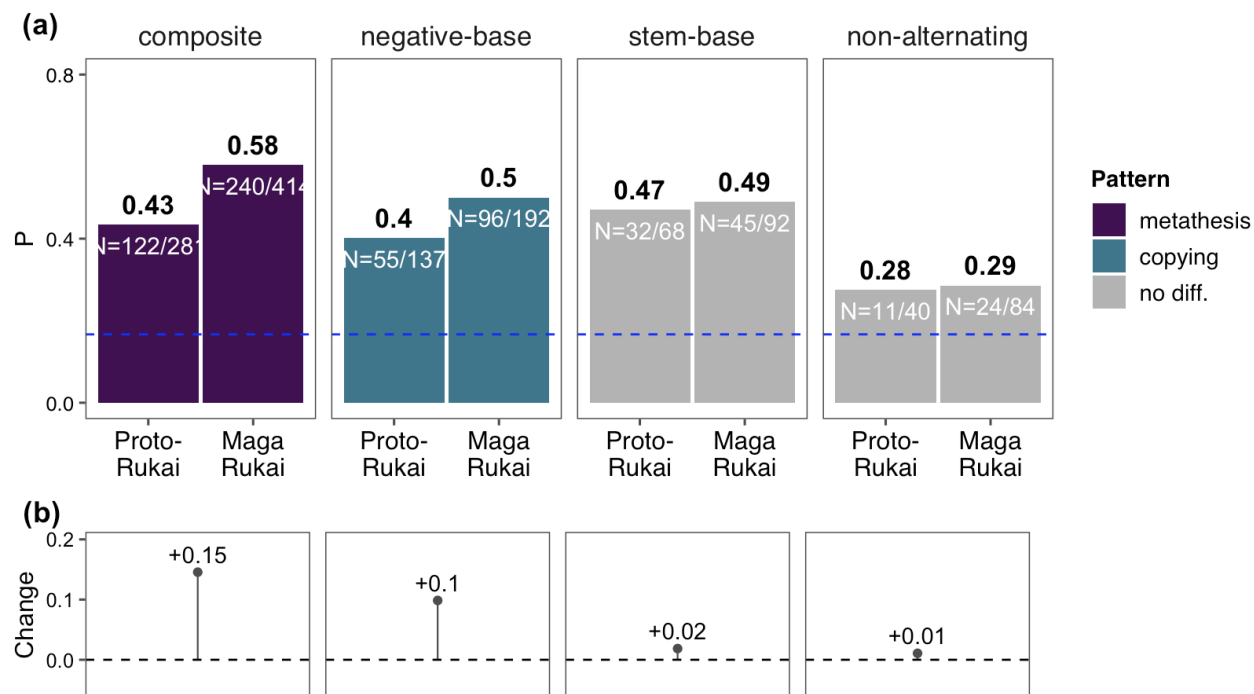


Figure 1: Vowel-matching in Proto-Rukai vs. Maga-Rukai (between V1 and V2). (a) shows the proportions of vowel-matching, while (b) shows the change in proportion of vowel-matching between Proto-Rukai and Maga ($P_{Maga} - P_{PrRu}$). Dashed line indicates the chance-level rate of vowel-matching alternations.

Going into the results, Fig. 1 compares vowel-matching rates in Proto-Rukai vs. Maga Rukai, by the type of UR that each paradigm has. We see that for the composite and negative-base paradigms, there has been an increase in the proportion of vowel-matching forms. In particular this increase is the largest for the composite UR forms. On the other hand, stem-base and non-alternating paradigms have seen a near-zero increase in vowel-matching. These results support

the view that restructuring of UR vowels has occurred specifically to remove abstract URs, and in such a way that makes paradigms predictable from the stem allomorph.

It should also be noted that vowel-matching appears to already have been present as a weaker tendency in Proto-Rukai. Maga has 6-7 phonemic vowels (depending on whether /ə/ is treated as phonemic), meaning that if vowel-matching occurred at chance, it should occur about 16.7% (1/6) of the time; this value is shown for reference in Fig. 1. Even in Proto-Rukai, vowel-matching happened at well above this rate. This suggests that when faced with resolving the ambiguity caused by syncope, speakers utilized a tendency that was already present in the lexicon, causing it to become stronger over time.

5.6 Alternative accounts to abstractness-avoidance

So far, I have argued that patterns of vowel-matching and leveling both serve to remove abstract URs. In this section, I consider several alternative explanations for the observed patterns, and show that they do not provide a satisfactory account of the data.

5.6.1 Syncope as the innovative variant

Chen (2008) puts forth the idea that syncope is a innovative and recent development, and that paradigms which fail to undergo syncope actually reflect an earlier stage of Maga. Under this account, a paradigm like [událi]~[i-událí:] (cf. expected negative form [i-udlí:]) has not leveled towards the stem. Instead, the negative allomorph has simply failed to undergo syncope, and reflects a more conservative stage of the language. If we adopt this idea, then around 75% of the leveling cases in Section 5.1 Table (22) can be re-interpreted as forms that reflect a conservative stage of Maga, and have not yet undergone syncope.

This account is less satisfactory for several reasons. First, it does not explain leveling in composite-UR paradigms (e.g. [blávní:]~[i-k-blavní:]). For these, the leveled form has already undergone syncope, and cannot reflect an earlier pre-syncope stage of Maga. Additionally, in Section 5.1, leveling was shown to happen asymmetrically, happening at much higher levels for stem-base and negative-base paradigms than for composite-UR paradigms. If non-syncope is argued to reflect a more conservative stage of Maga, these asymmetries cannot be explained.

5.6.2 Historical explanations for failure to undergo syncope

Vowels that were historically long in Proto-Rukai have resisted syncope in Maga. As a result, apparent exceptions to the syncope pattern may actually be an artifact of historic long vowels, rather than the result of base-driven leveling. Proto-Rukai had diphthongs which monophthongized in Maga (e.g. *likolaw → rkúlo); more examples are given above in Section 5.4.1. In addition, Proto-Rukai *ʔ was uniformly deleted in Maga, and as a result *VʔV sequences became either a short monophthong V or bimoraic V: (e.g. *abonoʔo → abúnu ‘ant’). Both diphthongs and *VʔV sequences resisted syncope, as shown by the representative examples in (34).

Of the Maga Rukai words where syncope failed to apply *and* we have available protoforms, historic long vowels blocked syncope in just 39/109 stems and 4/6 negative forms (N = 43/115, Prop = 0.37).⁶ In other words, most cases of exceptional non-syncope *cannot* be traced back to historic long vowels. The majority of non-syncope cases look like (35), where the non-deleted vowel corresponds to a historic short vowel. More generally, trisyllabic forms like *daʔanə > [*dani] are quite rare in Maga, as most Austronesian words are disyllabic (Blust 2013), and are unlikely to account for a large proportion of the unlevelled paradigms.

(34) *Maga reflexes of Proto-Rukai diphthongs and *VʔV*

Proto-Rukai	Maga	Gloss
*daʔanə	dáni (cf. *dni:)	‘house’
*kawmo	kómu (cf. *kmu:)	‘Japanese (people)’
*bəʔəkə	bíki (cf. *bki:)	‘pig’

(35) *Examples of Proto-Rukai short vowels that fail to undergo regular syncope*

Proto-Rukai	Maga	Expected Maga	Gloss
*dokoʔo	dúku	d_kú:	‘grow (plant)’
*pəkə	píki	p_kí:	‘house lizard’
*ma-rimóro	ma-rímúru	ma-r_múru	‘forget’

5.7 Picture so far

Examination of Maga paradigms (and comparison of these paradigms with Proto-Rukai) suggests that several types of restructuring have occurred, collectively moving the language away from a rhythmic syncope analysis, and towards a more concrete grammar.

Looking first at the Hsin corpus (without yet considering comparison with Proto-Rukai), I find that many paradigms have been leveled or partially leveled. Negative-base paradigms have generally leveled towards the negative allomorph, while stem-base paradigms have leveled towards the stem. Composite-UR forms have leveled in both directions (with a preference for leveling towards the stem). Paradigms can also be categorized by their predictability from the stem allomorph; stem-base paradigms are always predictable from the the stem. Additionally, vowel-matching renders composite-UR paradigms predictable (via metathesis), and negative-base forms predictable (via vowel-copying) from the surface stem.

Fig. 2 summarizes these patterns of leveling. To account for effects of vowel-matching on patterns of leveling, in Fig. 2, composite and negative-base paradigms are also organized by whether or not the paradigm is vowel-matching. When the data is organized this way, it is evident that for paradigms that are both composite-UR and vowel-matching, there has been relatively

⁶This count does not include Proto-Rukai forms where the long vowel is not in a syncope position, e.g. *capəʔə → cpí:.

less leveling than other paradigms. These paradigms can be analyzed as undergoing metathesis, making them concrete.

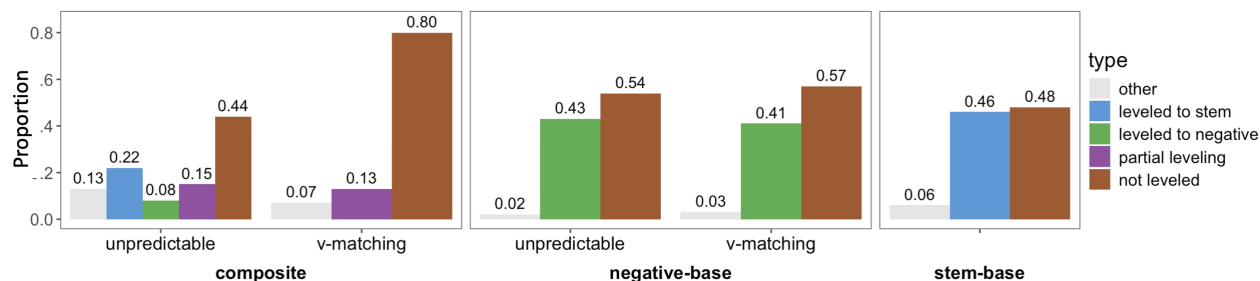


Figure 2: Summary of patterns of restructuring in Hsin's corpus. Each bar is labeled with the proportion of items that category.

Comparison of Proto-Rukai and Maga Rukai suggests that speakers have extended vowel-matching changes to composite-UR and negative-base paradigms; again, vowel-matching increases these paradigms' predictability from the stem. It is difficult to directly compare the patterns of leveling with these vowel-matching changes, as the former is based on patterns of syncope in the modern Maga lexicon, while the latter relies on comparison of Proto-Rukai and Maga Rukai. Nevertheless, we can approximate the proportion of paradigms that were restructured to result in metathesis or copying. Based on Section 5.5 (Fig. 1), around 17% of composite-UR forms were changed to be vowel-matching. Working backwards, this means that in Hsin's corpus, around 17% of the metathesis-compatible paradigms were originally unpredictable from stem, but have been restructured to be analyzable as metathesis. Similarly, around 10% of negative-base have been restructured to be vowel-matching, making them amenable to a concrete vowel-copying analysis. Fig. 3 summarizes the patterns of restructuring in Maga after accounting for vowel-matching changes.

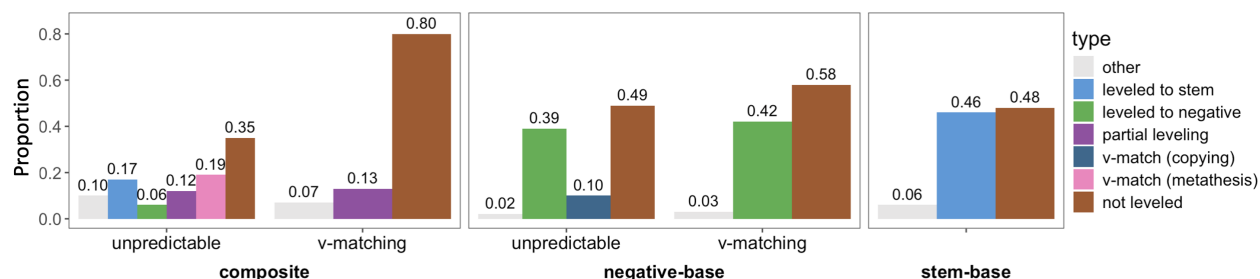


Figure 3: Summary of patterns of restructuring in Hsin's corpus, including vowel-matching changes

Overall, patterns of restructuring in Maga support the view that learners are biased against learning abstract URs. Leveling out alternations removes the need for abstract URs. Similarly, vowel-matching renders paradigms predictable from both surface allomorphs, removing the need for abstract URs. More generally, there is a tendency for restructuring to be based on the stem allomorph, but speakers must also have access to both the stem and negative allomorph, as leveling

happens in both directions. This is in line with the KK-C (choosing from allomorphs) and KK-C' (listed-allomorphs) levels of UR learning.

6 A surface-oriented model of Maga vowel alternations

In this section, I propose a formal analysis of Maga alternations which is compatible with a concrete-UR analysis. This analysis is based in parallel Optimality Theory. Additionally, I analyze alternations as morphologically driven; in other words, learners have turned what used to be phonologically-motivated syncope into morphologically-conditioned root alternation.

In Section 6.1, I first summarize some key characteristics of the Maga alternation pattern that make it difficult to account for in purely phonological accounts. In Section 6.2, I walk through two alternate abstract analyses, including Hsin's (2000) analysis of Maga alternations and a closely related alternative analysis based in stratal OT. These analyses are shown to either not work, or not predict the observed patterns of restructuring.

Then, in Section 6.3, I introduce a concrete analysis of Maga alternations based in OT. As will be shown, this analysis utilizes indexed constraints, and requires relaxing restrictions around the locality of constraint indexation. Crucially, this analysis can explain the observed directions of restructuring and correctly predict both leveling and vowel-matching. Finally, in Section 6.6, I also implement this analysis in Maximum Entropy Harmonic Grammar (MaxEnt; Goldwater & Johnson 2003), a probabilistic variant of Optimality Theory which uses weighted (instead of ranked) constraints and generates a probability distribution over the set of candidate outputs. Using this MaxEnt model, I compare several models that differ in their permitted level of UR abstractness, and find that the 'choose among allomorphs' (KK-C) and 'listed allomorphs' approaches outperform competing models in predicting patterns of restructuring in Maga.

6.1 Challenges to a phonological account of Maga alternations

The first challenge to a purely phonological account of Maga vowel alternations is that the domain of vowel alternations is sensitive to morphological structure, and not purely phonological. In particular, under a traditional rhythmic syncope analysis, odd-numbered vowels of monomorphemic words delete (e.g. //ḁ₁ma₂ri// → [ḁ₁ma₂ri]). In affixed words, however, it is the even-numbered vowels that delete (e.g. //i₁-ka₂-ḁ₃ma₄ri₅-V// → [i₁-k₂-ḁ₃m₄ri₅]).

Relatedly, as described in Section 2.2, affixes can also be split into alternating and non-alternating; Table (36), repeated from Table 8, lists examples of each affix type. There are no unifying phonological properties that separate these two groups. Non-alternating affix always appears closer to the stem (e.g. [i-s-kiɲi:] 'neg-to.put.on-clothes'), and can change word category and meaning. Based on these observations, the alternating affixes may broadly be characterized as inflectional, while the non-alternating affixes are derivational.

(36) *Examples of alternating vs. non-alternating affixes*

Type	Affix	Affix gloss	Example	Gloss
Non-alternating	ma-	stative vb.	ma-rkími	‘surprised’
	u-	active vb.	u-θlíbi	‘to fly’
	si-	to put on	si-kpíŋi	‘to put on clothes’
Alternating	i- -V	negative	i-θílbí:	‘to not fly’
	-a:	imperative	sukl-a:	‘Pick up! (imperative)’
	l-	plural	l-valki	‘children’

Second, as described in Section 5.2, patterns of partial leveling in longer words suggest that speakers may have learned an alternation affecting just the left edge of the root. This is important to note, as rhythmic syncope patterns are commonly analyzed as resulting from general constraints on metrical structure, which do not straightforwardly predict patterns of partial leveling.

The final key characteristic of the Maga alternation pattern is that there is surface non-locality in how Maga affixes interact with vowel alternations in the root. In particular, the imperative /-a:/ productively triggers alternations (see examples in (37)), but is a suffix and therefore non-adjacent to alternations at the left edge of the root. This non-locality is not an issue for some analyses of the Maga pattern, including Hsin’s analysis that is laid out below in Section 6.2. However, it is challenging to account for if learners have learned a left-edge alternation, as suggested by the partial leveling cases.

(37) *Nonlocality in affix-root interactions*

- [u-skúlu] ~ [sukl-á:] ‘pick up (imperative)’
[u-θlíbi] ~ [θílb-á:] ‘fly (imperative)’

6.2 Alternative analyses that utilize abstract URs

In this section, I consider alternate analyses of the Maga vowel alternations which stratify phonology across morphological domains and utilize abstract URs. I first briefly discuss a Lexical and Stratal Phonology account of Maga alternations, and argue that independent of issues around UR abstractness, such an account cannot capture empirical facts about the Maga alternation pattern. The second analysis, adapted from Hsin (2000), makes use of cyclic application of phonology. While this approach can capture the Maga rhythmic syncope pattern, it requires special reference to morpheme identity. This supports the idea that Maga vowel alternations are morphologically motivated. Crucially, both analyses also assume abstract URs, and therefore do not predict the patterns of restructuring that are observed in the data.

6.2.1 Metrical analysis in a stratal framework

Stratal approaches such as Lexical Phonology (Kiparsky 1982, Mohanan 1982) and Stratal OT (Bermúdez-Otero 1999, Kiparsky et al. 2000), which interleave morphology and phonology, offer a potential solution to the morpheme-specific nature of Maga alternations, as these approaches predict a split in the behavior of inflectional and derivational affixes. Stratal approaches can also account for issues of metrical incoherence; recall that Maga syncope appears to be conditioned on left-aligned iambic feet, while stress is conditioned on right-aligned trochaic feet. This is challenging to account for in parallel approaches like classical OT, but falls out straightforwardly in stratal OT if the two types of metrical structure are assigned at different levels of the grammar. In fact, Kaplan (2022) analyzes Southern Pomo rhythmic syncope, which is descriptively very similar to Maga, in Stratal OT. However, a stratal/Lexical Phonology analysis actually does not easily capture the inflectional/derivational split in Maga.

In Lexical Phonology, the distinction between affixes that do and don't trigger syncope can be captured by interleaving rules of morphology among rules of stem-level phonology. In particular, some affixes are assigned prior to stress assignment; these shift the position of metrically strong vowels, and therefore shift the position of syncope vowels. Affixes ordered after stress assignment will not trigger stress (and related segmental) alternations. Under this approach, derivational affixes that occur closer to the root should trigger vowel alternations (as they are ordered before stress assignment), while inflectional affixes should not. However, the Maga data does not match these predictions; as shown by the examples in (38), it is the inflectional affixes that trigger vowel alternations compared to the bare root, which is shown here for reference.

(38) *Examples of affixed forms*

Word type	deriv + root	inflec + root	deriv + root	inflec + deriv + root
Alternates?	no	yes	no	yes
Bare root	cɲíli	cɲíli	kpiɲi	kpiɲi
Example	u-cɲíli	i-ciɲlí:	si-kpíɲi:	i-s-kipíɲí:
Gloss	'see'	'not see'	'to put on clothes'	'to not put on clothes'

6.2.2 Cyclic application of phonology

Hsin (2000) circumvents the ordering contradictions that arise in Lexical Phonology by adopting the cyclic phonology framework of Halle & Vergnaud (1987). In this framework, word-level phonology consists of a cyclic and non-cyclic block. Affixes are marked (by diacritics) as either cyclic or non-cyclic, and each cyclic affix initiates a pass through the cyclic block of rules. After this, the entire word is subjected once to the non-cyclic block. Hsin proposes that cyclic phonology assigns left-aligned iambic feet. In the non-cyclic block, syncope applies followed by penult stress assignment. Affixes are cyclic, triggering both erasure of existing metrical structure

and reassignment of iambic feet. I will argue that while Hsin’s analysis works, it requires several stipulations and special reference to morpheme identity.

I assume, following Hsin, that negated verbs include verb-marking prefixes in the underlying form; for example, the UR of [i-ciŋlǐ:] ‘not see’ is /i-u-ciŋili-V/. The derivations in (39), adapted from Hsin (2000), illustrate the cyclic analysis; brackets are used to show the internal morpheme structure at each stage of the derivation. I also use foot-based representations for simplicity, although Hsin actually assigns stress using Halle and Idsardi’s 1995 grid-based algorithm.

Broadly speaking, each affix erases existing metrical and syntactic structure, and triggers reassignment of metrical structure. Additionally, in the cyclic phase, an affix forms its own prosodic unit, and is not metrified with the stem.⁷ For the paradigm [u-cŋǐli]~[i-ciŋlǐ:], in the first cycle, left-aligned iambs are built on the innermost constituent (the root). Then, in (39b), addition of the negative affix triggers another pass through the cyclic block; crucially, this pass also erases the boundary between the inner affix /u-/ and the root. As a result, [u-] forms an iambic foot with the initial syllable of the root. Subsequent application of rules in the non-cyclic block yields the output form [i-ciŋlǐ:]; here, the /u-/ prefix is deleted because it is in a weak position. The paradigm [sikpǐŋi]~[iskipŋi:] can be derived in a similar way.

(39) *Cyclic derivation of vowel alternations*

	a. /ciŋili/ ‘see’	b. /ciŋili/ ‘not see’	c. /kipiŋi/ ‘to put on clothes’	d. /kipiŋi/ ‘to not put on clothes’
Cyclic block				
Cycle 1	[u-[ciŋili]]	[i-[u-[ciŋili]]-V]	[si-[kipiŋi]]	[i-[si-[kipiŋi]]-V]
Left-aligned iambs	(u-)(ciŋi)(li)	i-(u-)(ciŋi)(li)-V	(si-)(kpi)(ŋi)	i-(si-)(kpi)(ŋi)-V
Cycle 2	–	[i-[uciŋili]-V]	–	[i-[sikiŋi]-V]
Left-aligned iambs		(i-)(uci)(ŋili)(-V)		(i-)(siki)(piŋi:)
Non-cyclic block				
Syncope	(u)(c_ŋi)(li)	(i)(_ci)(ŋ_li)(V)	(si)(k_pi)(ŋi)	(i)(s_ki)(p_ŋi:)
Right-aligned trochees	(u)(cŋǐ.li)	(i.ci)(ŋlǐ:)	(si)(kpi.ŋi)	(i.ski)(pŋi:)
SRs	[ucŋǐli]	[iciŋlǐ:]	[sikpiŋi]	[iskipŋi:]

Notably, this analysis only works if verb marking prefixes undergo syncope, even when they are onsetless. This contradicts evidence that in general, onsetless vowels do not delete (e.g. [uváci] ‘vein’, [lθiéki] ‘waterfall’; see Section 2.3). Hsin’s analysis therefore requires stipulations about which vowels can and cannot undergo syncope; in other words, syncope is still morpheme-specific and cannot be analyzed by general metrical constraints. More generally, this analysis, while able

⁷Hsin makes the distinction between cyclic and non-cyclic prefixes, but that is actually not necessary for this analysis to work; the distinction between alternating and non-alternating affixes is a direct consequence of morpheme order.

to account for regularly alternating paradigms, makes no specific predictions about how restructuring will occur. First, it assumes abstract URs, so does not predict vowel-matching changes or leveling to happen. Partial leveling is also not predicted to happen under this analysis, as syncope falls out from iambic stress assignment that applies throughout the word.

6.3 Concrete analysis of Maga alternations

In this section, I introduce an analysis of Maga vowel alternations that is compatible with concrete URs, and show that it predicts the observed patterns of restructuring. Constraints will be explicated using classical OT for ease of interpretation. In Section 6.6 below, I will then implement the proposed constraint set in a probabilistic weighted constraints framework, and compare several models that differ in their permitted level of UR-abstractness.

To illustrate constraint interaction, I (temporarily) assume that the UR corresponds to the stem allomorph; for example, given a paradigm [ɕɲíli]~[i-ɕɲíli:], the stem UR is /ɕɲíli/ ‘see’. I also follow Hsin (2000) in several analytical choices. First, final lengthening is assumed to be either allophonic or a morphological exponent of negation, and not part of a root’s underlying representation. Additionally, verb-marking prefixes are part of the input to negated forms (e.g. /i-ka-ɕmari-V/ ‘not moon’, /i-u-ɕɲíli-V/ ‘not see’). Finally, following Hsin, I assume that Maga allows complex onsets up to two consonants long, and no codas; in all the following tableau, consonant clusters are parsed as onsets (e.g. [blá.vni], not *[bláv.ni] ‘silver’).

In this section, I first introduce an indexed constraint and show that it can account for deletion of verbal prefixes in the negative form (and with other alternation-triggering affixes). I then show that the same constraints can explain alternations of the root, and predict both vowel-matching alternations and paradigm leveling.

6.3.1 Deletion of verbal prefixes

First, morpheme-specific behavior can be accounted for by adopting indexed constraints (Pater 2007). Under this approach, any constraint in a language’s grammar may be indexed to specific morphemes or sets of morphemes, capturing the idea that some alternations are exceptionally triggered by morphemes, rather than motivated by general phonology. The indexed constraint that I propose is adapted from Chen (2008), who analyzes syncope of Maga monomorphemes in OT. Chen motivates syncope using the constraint ALIGN(V-R-Ft) (align parsed vowels with the left edge of a foot). This constraint captures the intuition that in unaffixed stems like [ɕmári] ‘moon’, vowels tend to be aligned to the right of the word. However, negative forms (and more generally words with an alternating affix) show the opposite pattern, where root vowels tend to surface closer to the left edge of the word (e.g. [i-k-ɕamrí:] ‘not moon’). To capture this observation, I introduce the constraint Align(I, V_{Re}), defined in (40). This constraint requires that a prefix indexed *i* be aligned with a root vowel. Note that unlike Chen’s constraint, which aligns vowels with a word edge, this constraint simply requires that any root vowel be aligned with a prefix.

As will be shown below, this constraint formulation is able to capture patterns of partial leveling. Existing work has used similar alignment constraints (John J McCarthy & A. Prince 1993), which require alignment of the a morpheme with a phonological feature (e.g. Kager 1996).

- (40) *ALIGN(I, V_{Rt})*: Assign one violation mark for each segment which intervenes between the left edge of a morpheme indexed I and a vowel of the root.

Tableau (41) demonstrates how *ALIGN(I, V_{Rt})* can be used to derive deletion of verbal prefixes. First, in negative forms of active verbs, the active verb prefix should delete, as in /i-u-ara-v/ → [i-ará:] ‘not obtain’. In the faithful candidate (a), /u-/ intervenes between the root and /i_I/, resulting in a fatal violation of *ALIGN(I, V_{Rt})*. The winning candidate (b) removes this violation by deleting the active prefix.

In negative forms that take the stative verb prefix /ka-/, the vowel deletes but [k-] remains, as in /i-ka-isi-V/ → [i-k-isi] ‘urine’. This falls out from a ranking of IO-MAX-C >> *ALIGN(I, V_{Rt})*. The faithful candidate (c), where the prefix has not deleted, is again ruled out because it has the most violations of *ALIGN(I, V_{Rt})*. Candidate (e) fully deletes the stative prefix and has no violations of *ALIGN(I, V_{Rt})*, but is ruled out because it violates a higher-ranked faithfulness constraint IO-Max-C. As a result, (d) emerges as the winner.

(41) *Tableau: deletion of verbal prefixes*

/i _I -u-ara-V/	IO-MAX-C	<i>ALIGN(I, V_{Rt})</i>	IO-MAX-V
a. i _I -u-ara:		1!	
☞ b. i _I -__-ara: (delete u)			1
/i-ka-isi-V/			
c. i _I -ka-isi:		2!	
☞ d. i _I -k__-isi: (delete a)		1	1
e. i _I -__-isi: (delete ka)	1!		1

Notably, Pater (2007) explicitly limits the scope of indexed constraints so that the locus of constraint violation is within the indexed morpheme. This locality requirement makes it difficult to account for the Maga alternation patterns; under the current analysis, the imperative suffix /-a:/ would have to trigger vowel alternations at the left edge of the word, but the suffix itself is at the right edge of the word. To account for the Maga pattern, locality requirements around indexed constraints must be relaxed. In particular, I assume that in words where the outermost affix is indexed to I, the entire word (i.e. affix and derivational base) is also indexed to I. Additionally, the alignment constraint is itself modified, and restated as in (42). The new constraint, *ALIGN-L(W_I, V_{Rt})*, requires that a root vowel be indexed not to an alternating affix, but instead to the left edge of a *word* indexed to I. A similar approach is proposed by Jurgec & Bjorkman (2018), who show that relaxing the locality restriction of indexed constraints is necessary for capturing certain non-local derived environment effects.

- (42) *ALIGN-L(W_I, V_{Rt}): Assign one violation mark for each segment which intervenes between the left edge of a word indexed I and a vowel of the root.*

Tableau (44) reanalyzes /i-u-ara-V/ ‘not obtain’ using the new *ALIGN-L(W_I, V_{Rt})* constraint. In this tableau, entire candidate words are indexed to I. As before, in candidate (b), the active prefix is deleted to reduce violations of *ALIGN-L(W_I, V_{Rt})*. Notably, this new constraint aligns the left edge of the *word*, not just the negative prefix, to a root vowel. As such, deletion of the negative prefix /i-/ could happen to further reduce violations of *ALIGN(I, V_{Rt})*, as in candidate (c). This candidate is presumably ruled out by a morpheme realization constraint (Kurusu 2001), which requires that inflectional morphemes have an exponent in the surface form.

- (43) *Tableau: deletion of verbal prefixes with ALIGN-L(W_I, V_{Rt})*

/i _I -u-ara-V/	REALIZEMORPH _{INFLEC}	ALIGN(I, V _{Rt})	IO-Max-V
a. [i _I -u-ará:] _I		2!	
 b. [i _I -__-ará:] _I (delete u)		1	1
c. [__-__-ará:] _I (delete i, u)	1!		1

Tableau (44) demonstrates how *ALIGN-L(W_I, V_{Rt})* can resolve issues of non-locality in imperative forms, using the input /iki-a:/ → [iky-á:] ‘trample crops (imperative)’. Because the entire word is indexed to I, candidate (a) incurs a fatal violation of *ALIGN-L(W_I, V_{Rt})*, and (b) emerges as the winner. Note that the imperative suffix causes deletion or gliding of root-final vowels, but accounting for this alternation is beyond the scope of the current paper.

- (44) *Tableau: deletion of verbal prefix in imperatives*

/u-iki-á:/	ALIGN-L(W _I , V _{Rt})	MAX-V
a. [u-iky-á:] _I	1!	
 b. [__-iky-á:] _I (delete u)		1

In the rest of this section, I illustrate how *ALIGN-L(W_I, V_{Rt})* can also motivate both root vowel alternations and patterns of leveling. The rest of this section will walk through how this constraint motivates both alternation and paradigm leveling in each of the common paradigm types (composite, stem-base, and negative-base). The main constraints that will be used in the analysis are defined in (45); these include *CCC, NOCODA (Zec 2007), and standard input-output correspondence constraints (John J. McCarthy & A. S. Prince 1995).

- (45) a. *CCC: Assign one violation for each triconsonantal onset/coda cluster.
b. NOCODA: Assign a violation for each coda
c. *[+LONG]: Assign a violation for each [+long] segment.
d. IO-MAX-C, IO-MAX-V: Assign a violation for every consonant/vowel deletion.

- e. IO-LINEARITY: Precedence relations should be preserved by the corresponding segments in the output (prohibits metathesis).
- f. IO-INTEGRITY: Nothing in the input can have >1 correspondent in the output (prohibits vowel-copying).

6.3.2 Composite UR paradigms

First, consider paradigms such as [u-cɲɪli] ~ [i-ciɲli:] /cɲili/ ‘to see’, where the root and presumed UR are of the shape CCVCV. In these paradigms, the negative form can be derived by metathesis at the left edge of the root. This is shown in tableau (46); indices have been added to alternating segments for clarity. First, for the input /u-cɲili/, the faithful candidate (a) does not violate ALIGN-L(W_I , V_{Rt}), and therefore wins. In contrast, given the input /i₁-u-cɲili-V/, the faithful candidate (d) incurs a fatal violation of ALIGN-L(W_I , V_{Rt}). In candidates (f-i), deletion of the verbal prefix [u-] reduces violations of ALIGN-L(W_I , V_{Rt}) by 1.

Of the remaining candidates, (f) wins by undergoing metathesis to remove one more violation of ALIGN-L(W_I , V_{Rt}). Candidate (g) undergoes metathesis twice to further reduce violations of ALIGN-L(W_I , V_{Rt}), but is ruled out because it violates a highly ranked *CCC constraint. This constraint is independently motivated; except in a few exceptional cases that resulted from paradigm leveling, triconsonantal clusters are not observed in Maga. Candidate (h), which undergoes vowel copying to reduce violations of ALIGN-L(W_I , V_{Rt}), is eliminated by ranking INTEGRITY above LINEARITY. Finally, candidate (i), which reduces violations of ALIGN-L(W_I , V_{Rt}) by deleting the root-initial consonant, is eliminated by highly-ranked MAX-C. Note that although it is not shown in the tableau, a candidate like [i-icɲilɛ:], which has removed violations of ALIGN-L(W_I , V_{Rt}) by copying a vowel into the left edge of the word, is also possible. This candidate could be ruled out by either a constraint against long-distance copying or a constraint that enforces faithfulness of segments at the edge of the word (e.g. ANCHOR).

(46) *Tableau: metathesis motivated by ALIGN(L, V_{Rt})*

			*CCC	Max-C	ALIGN-L(W _I , V _{Rt})	Max-V	INTEGRITY	LINEARITY
/cŋ _a i _b li/								
☞ a.	cŋ _a i _b li	(faithful)						
b.	ci _b .ŋ _a li	(metathesis)						1!
c.	ci _b .ŋ _a i _b .li	(copying)					1!	
/i _I -u-cŋ _a i _b li-V/								
d.	[i-u-cŋ _a i _b .lí:] _I				4!			
e.	[i-cŋ _a i _b .lí:] _I	(delete u)			3!	1		
☞ f.	[i-.ci _b .ŋ _a lí:] _I	(delete u, metathesis)			2	1		1
g.	[i-.i _b .cŋ _a lí:] _I	(delete u, metathesis x2)	1!		1	1		2
h.	[i-.ci _b .ŋ _a i _b .lí:] _I	(delete u, copy)			2	1	1!	
i.	[i-.i _b .ŋ _a lí:] _I	(delete u, delete c)		1!	1	1		1

Longer composite-UR paradigms of the shape CCVCCVCV appear to undergo two instances of vowel metathesis, as in /rk_ai_brk_ci_dri/ [rkirkíri]~[i-ri_bk_ari_dk_crí:]. For these forms, vowel metathesis at the left edge of the root is motivated by ALIGN-L(W_I, V_{Rt}), but other alternations happen simply to resolve violations of general markedness restrictions. This is shown in tableau (47) for [rkirkíri]~[i-k-rikrikí:] ‘tremble’.

For simplicity, in this tableau and future tableaux, I only show candidates where the verbal prefix has already undergone syncope. Given the input /i-u-rkirkiri-V/, candidate (b), which undergoes one metathesis to reduce violations of ALIGN-L(W_I, V_{Rt}), ends up with a triconsonantal cluster, and is therefore ruled out by a fatal violation of *CCC. Candidate (c) undergoes another metathesis to resolve this violation, and emerges as the winner. Note that candidate (d) copies the first vowel of the root, resolving violations of ALIGN-L(W_I, V_{Rt}) without creating additional violations of *CCC. This candidate is exactly what is observed in cases of partial leveling, and could win if the rankings of INTEGRITY and LINEARITY are reversed. In a probabilistic MaxEnt implementation, candidates like (d) do in fact get assigned some probability, in line with the fact that Hsin’s corpus contains partially leveled paradigms.

(47) *Tableau: two metatheses in longer words*

/i-u-rk _a i _b rk _c i _d ri-V/		*CCC	ALIGN-L(W _L , V _{Rt})	INTEGRITY	LINEARITY
a.	[i-rk _a i _b rk _c i _d rí:] _I (faithful)		3!		
b.	[i-ri _b k _a rk _c i _d rí:] _I (1x metathesis)	1!	2		1
☞ c.	[i-ri _b k _a ri _d k _c rí:] _I (2x metathesis)		2		2
d.	[i-ri _b k _a i _b rk _c i _d rí:] _I (1x copying, partial leveling)		2	1!	

The analysis so far crucially relies on the assumption that paradigms are vowel-matching. Exceptions to this, such as [tmúsu]~[i-k-timsú:] ‘salt’, cannot be accounted for if we assume a non-composite UR like /tmusu/. This is demonstrated in tableau (48); note that in this tableau and other tableaux where the grammar predicts an output that diverges from the regular alternation pattern, ‘⇐’ points to the observed output prior to restructuring, and ‘⇒’ points to the innovative restructured output. In this tableau, the attested candidate (c) must undergo both metathesis and a change in vowel quality (violating IDENT[front]). Candidate (c) is harmonically bounded by (b), which undergoes just metathesis, and therefore (b) will always be the winning candidate. This outcome is desirable, because it predicts paradigms to be mislearned in a way that results in vowel-matching, exactly as observed in the comparative data.

(48) *Tableau: predictions for non-matching paradigms*

/i _I -ka-tmusu-V/		ALIGN-L(W _L , V _{Rt})	LINEARITY	IDENT[front]
a.	[i-k-tmúsú:] _I	4!		
⇒ b.	[i-k-tumsú:] _I (metathesis)	3	1	
⇐ c.	[i-k-timsú:] _I (vowel change)	3	1	1!

6.3.3 Stem-base paradigms

Most stem-base paradigms have a stem of the shape VCVCV, such as [u-icíbi]~[i-icbí:] /icibi/ ‘bake in stone’. In regularly alternating paradigms, the negative form can be derived by vowel deletion (e.g. /i-u-icibi-V/ → [i-ic_bí:]). However, about half of these paradigms have leveled towards the stem form, as in [událi]~[i-udalé:] /udali/ ‘rain’. The grammar so far actually straightforwardly predicts leveling of stem-base paradigms; this is because given an input like /i-u-udali-V/, the initial root vowel is already as aligned to the left as it can be, and further changes (via metathesis or copying) would be spurious. This is shown in tableau (49); the faithful candidate (a) does not violate ALIGN-L(W_L, V_{Rt}) or other markedness constraints. As a result, any non-faithful candidates are harmonically bounded and will be ruled out.

(49) *Tableau: leveling in stem-base paradigms*

/i _I -u-udali-V/	ALIGN-L(W _I , V _{Rt})	MAX-V	LINEARITY
⇒ a. [i-ud _a ab _b lí:] _I		1	
⇐ b. [i-k-ud _a lí:] _I (deletion)		2!	
c. [i-k-ua _b d _a lí:] _I (metathesis)		1	1!

6.3.4 Negative-base paradigms

Most negative-base paradigms have stems of the shape CCV, as in [rná:] ~ [i-k-raná:] /rna/ ‘creek’. With these paradigms, the negative form can be derived via vowel-copying. This is demonstrated in tableau (50) for /i-ka-rana/. Metathesis in candidate (b) results in a final consonant; final consonants, and more generally codas, are never found in the Maga lexicon, so (b) can be ruled out by a highly ranked and independently-motivated NOCODA constraint.⁸ As a result, candidate (c), which undergoes vowel-copying, emerges as the winner.

(50) *Tableau: vowel-copying*

/i-ka-rna-V/	NOCODA	ALIGN-L(W _I , V _{Rt})	Integrity	Linearity
a. [i-k-rná:] _I (faithful)		3!		
b. [i-k-rán] _I (metathesis)	1!	2		1
☞ c. [i-k-raná:] _I (copying)		2	1	

Input stems of the shape CCV typically surface with lengthening of the final vowel, as in /rna/ → [rná:]. This can be attributed to a typologically common word minimality requirement that words be at least two moras (or syllables) long (Hayes 1995, Downing 2006). However, in a large proportion of these paradigms, the stem has also leveled to the negative allomorph, such as in ‘juice, sap’ [túru] ~ [i-k-turú:].

Tableau (51) illustrates how leveling of this type may arise. Given a pre-leveled input like /tru/, the faithful candidate (a) fatally violates a highly ranked word minimality requirement, enforced here using the constraint WORDMIN. To resolve this violation, outputs can either undergo final vowel lengthening, as in (b), or copy the stem vowel, as in (c). If INTEGRITY is ranked above IDENT[long], then the un-leveled candidate (b) will win. If, however, the rankings are reversed, then the leveled out candidate (c) will win.

(51) *Tableau: leveling to the negative form*

/tru/	WORDMIN	IDENT[long]	INTEGRITY
a. trú	!1		
⇐ b. trú: (lengthen)		1	
⇒ c. túru (copying, leveled)			1

⁸Alternatively, candidate (b) could be ruled out by a morpheme exponence constraint which requires the negative form to end in a final long vowel.

Note that in this account, leveling results from vowel-copying, and only works if the leveled form has matching vowels, as in [túru] ‘juice, sap’. However, we do see some leveled paradigms where this is not the case, such as [ma-pári]~[i-k-parí:] /pari/ ‘stinking’. This issue is addressed in Section 6.5 below.

6.4 Summary: predictions of the OT analysis

Table (52) summarizes predictions of the analysis laid out so far. Note that these predictions are based in strictly-ranked OT. However, they should align with the outputs that are predicted to have high probability in the MaxEnt implementation to be laid out in Section 6.3. Overall, this analysis is able to closely match the observed patterns of restructuring from Section 5. That is, for composite-UR and to some extent negative-base paradigms, we should see vowel-matching changes that affect the negative allomorph. Additionally, negative-base and stem-base paradigms should level towards their respective bases.

(52) *Summary of predictions by base type* (* = results in vowel-matching)

Type	Output: stem	Output: negative	Example
composite	faithful	metathesis*	[u-çŋili]~[i-çŋli:]
	faithful	metathesis*	[u-rkirkíri]~[i-rikrikrí:]
negative-base	copying (lv to neg)	copying*	[túru]~[i-k-turú:]
stem-base	faithful	faithful (lv to stem)	[událi]~[i-udalí:]

Though not described in detail here, regular non-application of syncope pattern can be accounted for using general markedness constraints such as *GEMINATE and *CCC, which hold true across the lexicon. For example, in the paradigm [tlalé:]~[i-k-talalé:] ‘chief’, the negative form does not undergo metathesis (cf. *[i-k-tallé:]). In this example, metathesis is ruled out by a highly ranked *GEMINATE constraint.

6.5 The space of possible URs

The analysis above explains the basic patterns of vowel alternations, but also assumes that the UR always corresponds to the stem allomorph. However, as discussed above, some paradigms level towards the negative form in ways that are not easily explainable in a vowel-copying analysis. This suggests that speakers also have access to the negative allomorph in UR-learning.

There are two ways of accounting for this, and I test both in the model comparisons below. One is that learners only learn one UR for each paradigm, but this UR can correspond to either the stem or negative forms, in line with a KK-C level of abstractness. To implement KK-C URs, I make the simplifying assumption that the learned UR will correspond to the surface form that preserves the most vowel contrasts, and otherwise default to the stem form. For example, for [rdé:]~[i-k-rudé:], the UR would be /rude/, corresponding to the negative allomorph. For [smítu]~[i-k-simtú:], both allomorphs are equally informative, so the UR is /smitu/. Future work should consider models of

UR learning that simultaneously learn the UR and constraint weights (for a review, see Wang & Hayes 2025).

Alternatively, learners could list multiple URs for each paradigm, each corresponding to a surface allomorph. I implement this using a base competition model (Breiss 2021, 2024), in which outputs are faithful to multiple listed allomorphs (i.e. bases). Following Breiss (2021, 2024), this model has two sets of faithfulness constraints, each enforcing faithfulness to their respective base. Additionally, as described in Section 6.6.2, stem-faithfulness constraints are biased to have a higher weight, explaining why learners generally restructure paradigms towards the stem allomorph. Another more nuanced approach, explored in Breiss (2024), would be to scale faithfulness to each base by their resting activation levels.

Tableau (53) shows a tableau for [u-cɨŋli] ~ [i-cɨŋli:] ‘see’, revised to reflect a listed allomorphs grammar. Crucially, there are now two versions of each faithfulness constraint, in this case LINEARITY, respectively enforcing faithfulness to the stem and negative allomorphs. In this example where the two faithfulness constraints are unranked with respect to each other, in the stem form, either candidates (a) or (b) can win. For the negative form, high-ranked ALIGN-L(W_I , V_{Rt}) causes candidate (d) to win.

(53) *Listed allomorphs grammar*

/u-cɨŋli/, /u-cɨŋli/	ALIGN-L(W_I , V_{Rt})	LINEARITY _{STEM}	LINEARITY _I
☞ a. u-cɨŋli			*
☞ b. u-cɨŋli		*	
/i-cɨŋli-V/, /i-cɨŋli-V/			
c. [i-cɨŋli:] _I	*!		*
☞ d. [i-cɨŋli:] _I		*	

6.6 MaxEnt model implementation

In this section, I introduce a MaxEnt implementation of the analysis described above. I compare four models which differ primarily in their UR type, and are otherwise nearly identical. The models are a STEM-BASE (KK-B) model where URs always correspond to the stem form, a MIXED-BASE (KK-C) model where URs correspond to either the stem or negative allomorph, a LISTED ALLOMORPHS (KK-C’) model where both the stem and negative allomorphs are inputs, and a COMPOSITE UR model which assumes the URs required by a traditional morphophonemic analysis of Maga.

As described above, this model is implemented in MaxEnt (Goldwater & Johnson 2003). The probabilistic nature of MaxEnt allows for better comparison of model results with the Maga lexicon, where patterns of restructuring are variable. Additionally, a bias is implemented as a Gaussian prior following Wilson’s 2006 methodology. This prior is used to enforce a bias towards stem-based restructuring.

In all models, constraint weights were learned using the R package `maxent.ot` (Mayer, Tan & Zuraw 2024). In this package, constraint optimization is done using the `OPTIM` function from the `Rcore` statistics library and restricted to finite, non-negative values.

6.6.1 Inputs and constraint set

The constraint set includes the constraints introduced in Section 6.3, as well as IDENT constraints for vowel quality features. For the LISTED ALLOMORPHS model, IO-Faithfulness constraints are replaced with the equivalent OO-Faithfulness constraints, enforcing both faithfulness to the stem form and negative form.

The input to the model is the set of 790 stems from the Hsin corpus, modified to reflect a stage of Rukai prior to significant restructuring. First, paradigms were modified to have the expected alternation pattern, had syncope applied regularly. The rate of vowel-matching in the input was also changed to be roughly the same across all stem shapes, and be at around 45%, which was the rate seen in Proto-Rukai. To simplify the training data, forms were pooled by their syllable structure, alternating vowel qualities, and type of prefix (/i-/ vs /i-k-/); for example, [d̪mári] ~ [i-k-d̪amrí:] and [rd̪ámi] ~ [i-k-rad̪mí:] are both represented as [CCaCi] ~ [i-k-CaCCi:].

Example candidates for the model are shown in (54). For both the unaffixed and negative-prefixed inputs, I include the following candidates: the stem allomorph, negative allomorph, and outputs that resolve ALIGN-L(W_L , V_{Rf}) via metathesis, copying, and vowel deletion. Candidates that undergo consonant epenthesis/deletion are not considered, and assumed to be ruled out by highly weighted faithfulness constraints DEP-C and MAX-C. For longer words, I also include candidates that undergo a multiple metathesis or copying to resolve violations of general markedness (e.g. *CCC), as shown for /i-k-tkasluḍu/ here.

(54) *Example Input-candidate pairs*

Inputs	/i-k-tmusu:/	/i-k-tkasluḍu-V/
Candidates:	i-k-tmusú: (faithful)	[i-k-tkasluḍú:] (faithful)
	i-k-tumsú: (metathesis)	[i-k-taksluḍú:] (1 x metathesis)
	i-k-tumusú: (copying)	[i-k-taksulḍú:] (2 x metathesis)
		[i-k-takasluḍú:] (copying)

6.6.2 Incorporating a soft markedness bias

In the MaxEnt models, a ‘prior’ term is implemented as a Gaussian distribution over each constraint weight. During constraint optimization, the model learns constraints in a way that both maximizes log-likelihood (i.e. fit to the training data) and minimize the prior. The prior, calculated as in (55), is defined in terms of a mean (μ) and standard deviation (σ). For each constraint, w is its learned weight, and μ can be thought of as the ‘preferred’ weight. The numerator of the prior term reflects how much the actual weight deviates from the preferred weight of each constraint,

and the penalty resulting from the bias term increases as constraint weights diverge from μ . Note that size of this penalty will increase with the square of the weight. The value of σ^2 determines how much effect the preferred weight (μ) has; lower values of σ^2 result in a smaller denominator, and therefore greater penalty for weights that deviate from their μ .

$$(55) \quad \sum_{i=1}^m \frac{(w_i - \mu_i)^2}{2\sigma^2}$$

In the STEM-BASE, MIXED-BASE, and COMPOSITE UR conditions, $\mu = 0$ for all constraints while $\sigma^2 = 100$; this gives the models a small penalty for non-zero weights on any constraints, which ameliorates the tendency for models to overfit (Martin, Ney & Zaplo 1999). In the LISTED ALLOMORPHS condition, μ is still zero for all constraints, but σ^2 is manipulated so that the model prioritizes faithfulness to the unaffixed stem over faithfulness to the negative allomorph (Wilson 2006). Specifically, all FAITH_I constraints have $\sigma^2 = 0.8^9$ (resulting in a strong pressure towards low weights), while all other constraints have $\sigma^2 = 10$.

6.7 MaxEnt model results and comparisons

All three models with more concrete URs outperformed the COMPOSITE-UR model at predicting the modern Maga patterns. This is seen in Table (56), which shows the log likelihood ($\hat{L}$) for each model when fit to the modern Maga data (i.e. the Hsin corpus). The LISTED ALLOMORPHS model performed the best, followed by the MIXED-BASE and STEM-BASE models.

(56) *Table: Log likelihood ($\hat{L}$) of model predictions fit to modern Maga*

Model	KK-level	$\hat{L}$
STEM-BASE	KK-B	-1343.10
MIXED-BASE	KK-C	-1295.08
LISTED ALLOMORPHS	KK-C'	-1126.85
COMPOSITE	KK-D	-1395.20

A more detailed examination of model predictions shows that the LISTED ALLOMORPHS model in particular makes predictions which qualitatively match the observed patterns of restructuring (see Fig. 3). In contrast, the COMPOSITE UR model predicts very little restructuring. Consider Fig. 4, which shows the predicted probabilities of the COMPOSITE UR and LISTED ALLOMORPHS models, organized by type of paradigm and type of restructuring. Table (57) below gives examples for each type of restructuring. The MIXED-BASE and STEM-BASE models made similar predictions to the LISTED ALLOMORPHS model, so are omitted here in the interest of space.

Looking at Fig.4a, the COMPOSITE UR model under-predicts the amount of restructuring. Additionally, it does not predict paradigms to undergo metathesis or leveling to the stem, though

⁹This value was determined by testing σ^2 values ranging from 0.4-5 in increments of 0.2 and selecting the best-fitting model.

these were the two most common types of restructuring found in Section 5. In contrast, looking at Fig.4b, the LISTED ALLOMORPHS model predicts exactly the main patterns of restructuring found in Maga: composite-UR paradigms are restructured to show a metathesis pattern, while negative-base and stem-base paradigms level towards their respective bases. In the negative-base paradigms, the model also predicts some vowel-copying changes, which is compatible with the descriptive results. The main shortcoming of the LISTED ALLOMORPHS model is that it predicts little to no leveling in the composite-UR paradigms, even though around 20% of these paradigms have been leveled in the corpus data.

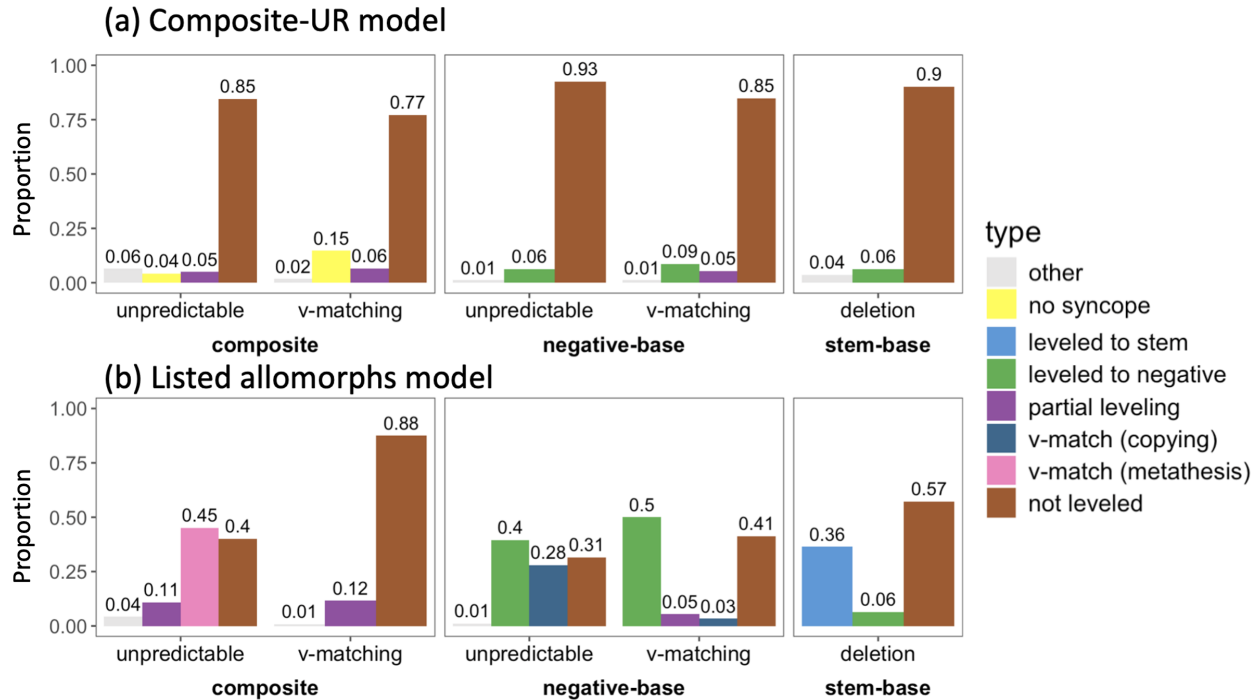


Figure 4: Predictions of the COMPOSITE UR and LISTED ALLOMORPHS models, by shape of input paradigm and type of restructuring

(57) *Examples of model predicted results*

Predicted change Example

no syncope	[plĩŋi~i-k-pilŋí:] → [piliŋi~i-k-piliŋí:]
level to stem	[u-ɕŋĩli~i-ciŋí:] → [u-ɕŋĩli~i-ɕŋĩli:]
level to negative	[rná:~i-k-raná:] → [rana~i-k-raná:]
partial leveling	[ma-θlímdi~i-k-θilimdí:] → [ma-θlímdi~i-k-θilimdí:]
copying	[ɖrími~i-k-ɖírmí:] → [ɖrimi~i-k-ɖirimí:]
metathesis	[tmúsu~i-k-timsú:] → [tmusu~i-k-tumsú:]

In sum, the COMPOSITE UR model performed worse than all other models at predicting the patterns of change found in modern Maga Rukai. In fact, even the most restrictive STEM-BASE model has a better fit to modern Maga (as measured by log-likelihood).

7 Discussion

7.1 Abstractness vs. generality in learning

In this paper, comparison of Proto-Rukai and Maga Rukai, and subsequent modeling in MaxEnt, suggests that an abstract-amenable rhythmic syncope alternation is being relearned as a more concrete system. In this new grammar, alternations are driven not by phonological restrictions, but rather by the morphology. In particular, I analyze alternations as motivated by an indexed constraint that targets the left edge of the root, rather than metrically-driven syncope. The new grammar maintains the surface alternations to some degree, but imperfectly accounts for the data, and as a result predicts substantial restructuring.

The results from Maga suggest that learners would rather learn more concrete representations, even at the cost of generality in the resulting grammar. This supports a growing body of work which finds that across languages, learners often acquire more concrete representations, even if doing so adds exceptionality and complexity to the phonological grammar. For example, Gouskova (2012) shows that Russian $\emptyset \sim V$ alternations (also known as *yers*), which were traditionally analyzed using abstract representational accounts, are better characterized with morpheme-specific constraints; in Polish, a similar set of *jer* alternations have evolved into a concrete system that is partially predictable from sonority sequencing (Czaykowska-Higgins 1988, Jarosz 2008, Rysling 2016). In Yidj, metrically-driven vowel alternations resulted in an abstract-requiring analysis, which was subsequently restructured into surface-predictable alternations that are productive, but not motivated by language-general phonology (Hayes 1999).

Notably, restructuring has not happened across-the-board; a portion of the lexicon (~40%) still appears to require abstract analyses, as the vowels of the negative form are not predictable from the stem, and vice versa (e.g. [tmúsu] ~ [i-k-timsú:]). There are several possible ways to explain the number of apparently abstract paradigms that remain in Maga. One possibility is that speakers are able to entertain abstract URs, but prefer not to; as a result, speakers will only infer abstract URs when there is strong evidence for them, such as when both allomorphs are highly frequent. The second possibility is that learners are memorizing surface forms, in line with the listed allomorphs approach (e.g. for a paradigm like [cɲili] ~ [i-ciɲlí:], they will learn two lexical representations, /cɲili/ and /ciɲli/). In this case, learners will only restructure forms that have no listed lexical entry, either because they have never learned the form before, or because it is low in frequency.

These two possibilities make similar predictions, where high-frequency paradigms can maintain apparently abstract patterns, while lower-frequency forms will tend to be restructured into more concrete paradigms. Although the lack of Maga language materials may make this difficult, future work should ideally test if these frequency-based predictions are borne out. More generally, the gradual and exceptional nature of Maga paradigm restructuring is not surprising; this is consistent with what has been observed for the restructuring of morphophonological paradigms in other languages. Such changes are more gradual and tend to be accompanied by lexical exceptions (e.g.

Harris 1989, Kiparsky 2003).

7.2 Restructuring as the result of language attrition

Much of the work on restructuring in rhythmic syncope languages has focused on speakers in the context of shift and attrition. Examples include Yidiñ (Hayes 1999), Nishnaabemwin (Valentine 2001), and Southern Pomo (Kaplan 2022). Likewise, Maga Rukai is spoken only by elderly members of the community who experience significant language contact with Japanese (the language of primary education in their youth) and Mandarin (the societal dominant language). This raises the issue that restructuring of rhythmic syncope may only arise in attrition contexts.

Even if this is the case, patterns of restructuring can still tell us about general biases in phonological learning. The instability of rhythmic syncope in attrition contexts suggests that even if abstract URs are learnable, they are only learned with extensive linguistic exposure that overrides a ‘default’ preference for concrete analyses. In addition, similar cases of restructuring have been found in commonly spoken languages that are not experiencing attrition. For example, Slavic yers involve a similar case of abstractness resulting from diachronic vowel deletion, which in many languages was then mislearned into a more concrete pattern.

Finally, languages described as having robust rhythmic syncope patterns have often not been scrutinized in detail. Macushi Carib and Southeastern Tepehuan are two languages described as having stable rhythmic syncope (Kager 1997). However, the Tepehuan pattern is potentially restricted to specific morphological contexts (E. Willett 1982, T. L. Willett 1989), and is based on a limited set of forms that has not been compared against more recent dictionary data (Willett & Willett 2015). For Macushi, the original report of rhythmic syncope came from Hawkins (1950), when the pattern had very recently developed. An examination of how the pattern has developed over time would be informative for understanding the stability of rhythmic syncope alternations.

7.3 Syncope as a morphologically-conditioned alternation

In this paper, $\emptyset \sim V$ vowel alternations were motivated by a constraint $\text{ALIGN-L}(W_I, V_{Rt})$, which is indexed to a set of (inflectional) affixes, references roots, and requires relaxing locality assumptions around indexed constraints. While these are non-standard assumptions, they make use of ingredients that are found in the morphophonology literature. First, it is not uncommon for specific morphemes to trigger alternations (Pater 2007). Second, there is reason to believe that learners are sensitive to roots and root-affix boundaries. Typologically, roots and affixes behave asymmetrically (Gouskova 2003); this has been formalized in the literature as root faithfulness (Beckman 1998) and markedness constraints on roots (Albright 2004). As discussed in Section 6, Jurgec & Bjorkman (2018) have also demonstrated a need for indexed constraints to relax locality restrictions.

More generally, in proposing $\text{ALIGN-L}(W_I, V_{Rt})$ as a constraint, I am analyzing $\emptyset \sim V$ alternations as morphologically-conditioned, rather than phonologically conditioned. The Maga pat-

tern could have started as phonetically-motivated vowel reduction, but then developed into a morphologically-conditioned alternation; this in line with findings that after initial phonologisation, an alternation will frequently narrow in its domain of application and eventually be lexicalized or morphologized (Bermúdez-Otero 1999). Additionally, my analysis is in line with prior work on diachronic restructuring, which have argued that alternations which arise from extensive restructuring are often not phonologically optimizing, and instead require constraints which explicitly require alternation (e.g. Anticorrespondence; Hayes 1999).

It could be that constraints like $\text{ALIGN-L}(W_I, V_{Rt})$ only arise in exceptional situations where learners are faced with a highly complex alternation. Hayes (1999) suggests that such constraints could be unstable and prone to being replaced with more general, phonologically optimizing grammars. In fact, there is some work suggesting that the next generation of Maga Rukai speakers (not the one documented by Hsin) have leveled out $\emptyset \sim V$ alternations even more extensively than in Hsin’s corpus (Author 2026). One reviewer also notes that this new generation has removed many consonant clusters via vowel epenthesis. These changes can be characterized as phonologically optimizing, and remove the need for an indexed constraint like $\text{ALIGN-L}(W_I, V_{Rt})$.

7.4 Placing Maga into the typology of metathesis and vowel-copying

Metathesis and vowel-copying in Maga are unusual because they do not follow typical cross-linguistic tendencies. Metathesis tends to be sporadic, phonetically incomplete, segmentally restricted to sonorants, and better characterized as gestural overlap than true segment transposition (Hall 2006, Mooney 2023). Similarly, copying tends to involve segmental restrictions, either of the vowels that can copy, or of the consonants that are transparent to copying (Mooney 2023). In contrast, metathesis and copying in Maga apply across a range of consonants with no apparent segmental restrictions. Fig. 5a shows, for the subset of metathesis-undergoing paradigms, the proportion of alternating forms. As seen here, the rate of alternation remains high regardless of the manner of the metathesizing consonant. Similarly, as shown in Fig. 5b-c, vowel copying is attested for most vowels, and across all manners of intervening consonants.

When metathesis and copying do occur with little to no phonological restrictions, they tend to be morphologically restricted (Mooney 2023). The Maga alternations fit into this characterization; they are non-phonologically optimizing and happen only with certain affixal triggers. In fact, the vowel alternations in Maga are in many ways similar to other cases of non-concatenative morphology, where morpheme exponents require roots to have a certain phonological form (John J. McCarthy & A. S. Prince 2017)

8 Conclusion

In Maga Rukai, a historical process of vowel deletion in prosodically weak positions resulted in a synchronic rhythmic syncope pattern, which can only be accounted for by positing abstract composite URs. In this paper, I propose that rather than maintaining rhythmic syncope, Maga

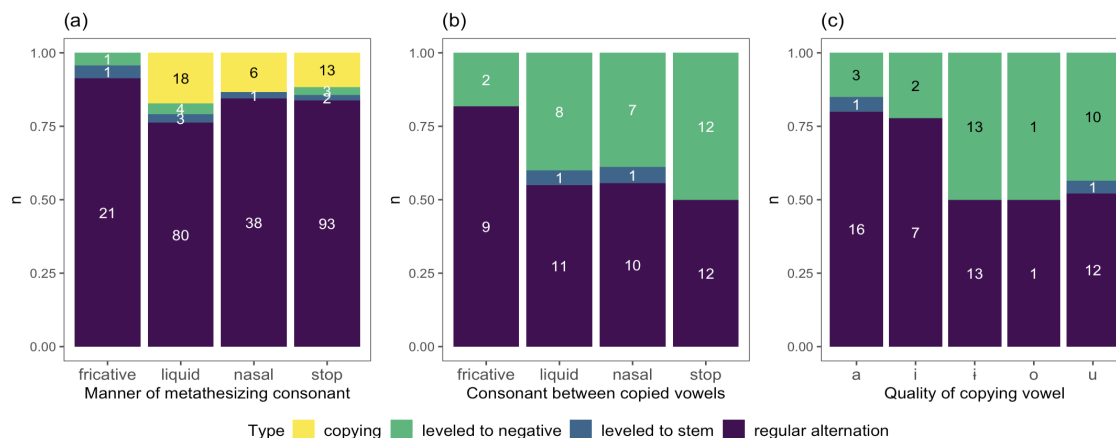


Figure 5: Rate of regularly alternating forms in paradigms that undergo (a) metathesis, by manner of metathesizing consonant; (b) copying, by manner of intervening consonant; (c) copying, by quality of copied vowel

paradigms were restructured in two ways. First, some paradigms preserved vowel alternations, but extended a vowel-matching pattern which made surface forms predictable from each other. Other paradigms leveled out all vowel alternations, where leveling was towards both the stem and negative allomorphs. Crucially, both types of restructuring make paradigms more concrete, removing the need for composite URs.

These results support a concrete view of UR learning, but it remains to be seen if abstract URs are always unlearnable, or if learners are simply biased against them. Notably, there are documented cases where abstract URs seem to be robustly learned. For example, Paramore & Bennett (2025) describe a nasal vowel alternation in Western Panjabi that appears productive, but requires positing abstract phonemes. It could be that UR-learning is ‘concrete by default’. That is, learners will assume that URs are concrete, and stop here if a concrete analysis accounts for some proportion of the data. It’s only when a concrete analysis falls below this ‘threshold accuracy’ that learners will consider more abstract URs. Future work should test this hypothesis, by comparing cases like Western Panjabi to Maga Rukai, with a focus on how well concrete alternative analyses account for the observed data.

In Maga, speakers leveled towards both the stem and negative allomorphs, which is compatible with either a KK-C (choosing from allomorphs) or KK-C’ (listed allomorphs) approach to UR-learning. Model comparisons (Section 6) suggest that of these two, the LISTED ALLOMORPH approach best accounts for the Maga patterns of restructuring. However, in the current study, UR-selection for the KK-C model was highly simplified (where the UR is whichever allomorph preserves the most vowel contrasts); future work should test the performance of a model that uses a more rigorous approach to UR learning.

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